RNA SYNTHESIS:

TRANSCRIPTION AND POSTTRANSCRIPTIONAL PROCESSINGS

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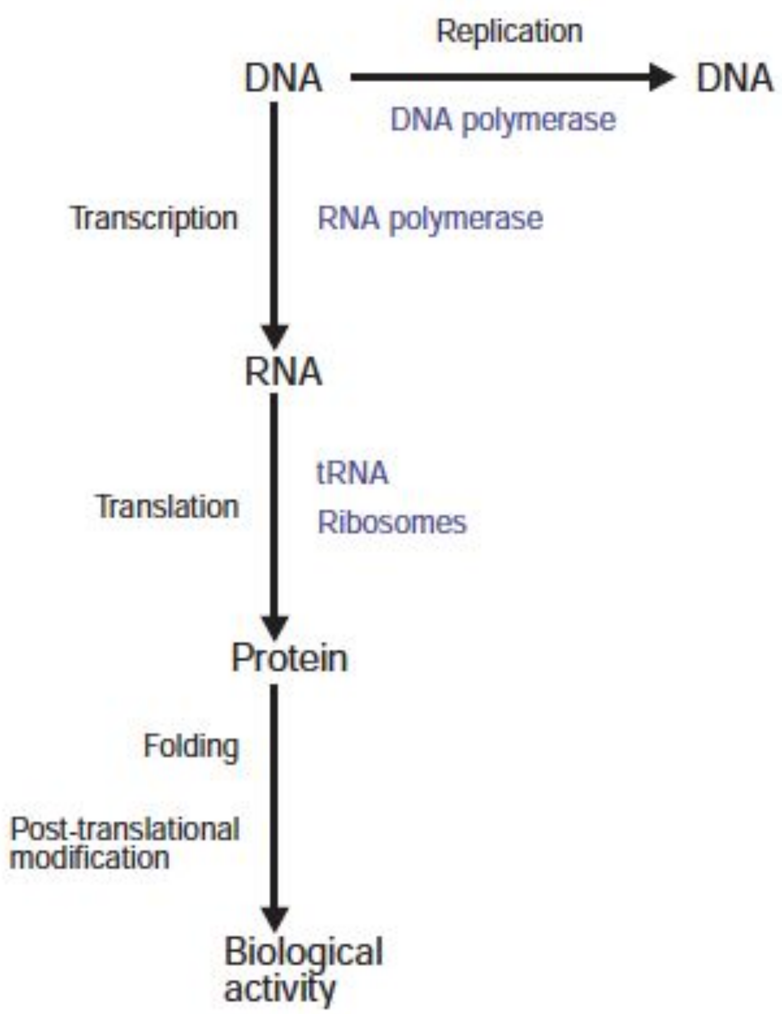
Objectives:

At the end of this chapter, it is expected;

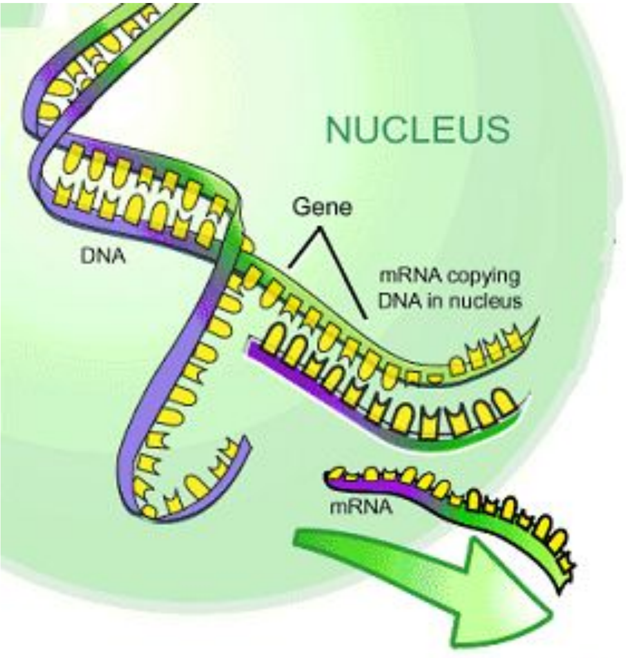
- To describe the features of transcription
- To identify enzymes and proteins involve in transcription and appreciate their roles.
- To describe the stages and mechanism of transcription
- To describe the mechanism of posttranscriptional processing
- To list the differences between transcription in prokaryotes and eukaryotes
- To identify the inhibitors of transcription
- To describe the process of Reverse transcription

Information flow

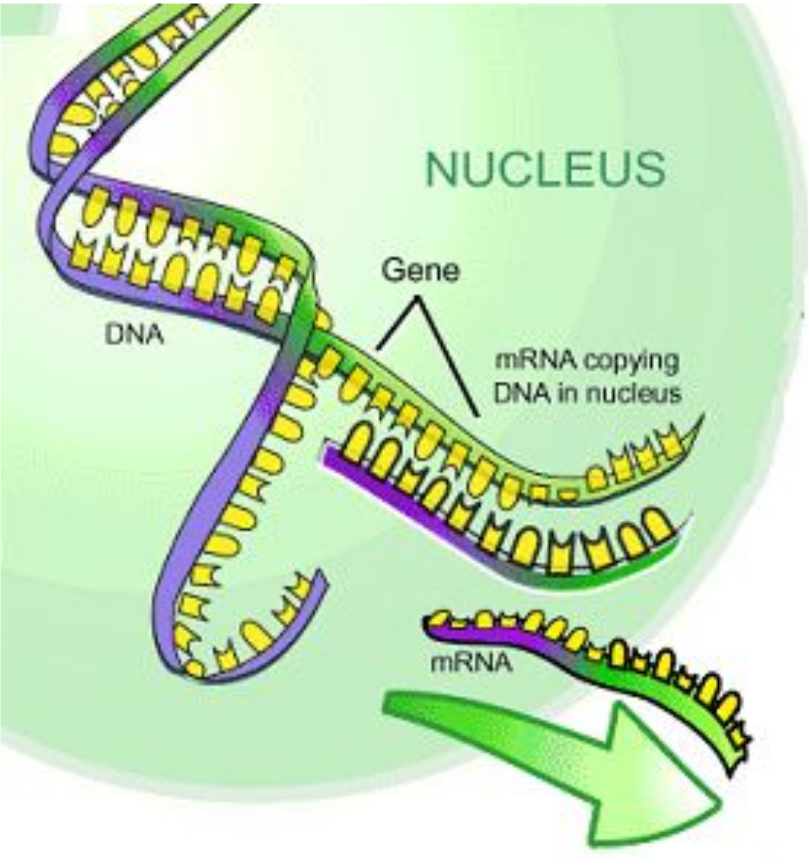


RNA synthesis: Overview

- Transferring the information in a DNA → RNA.
- Prokaryotes – Coupled T&T; mRNA, no modifications; rRNA & tRNA formed after post-translational processing.
- Eukaryotes - hnRNA - a primary transcript - further processed to remove introns.
    - rRNAs & tRNAs - the final product.
    - mRNA → translated to proteins.



Transcription



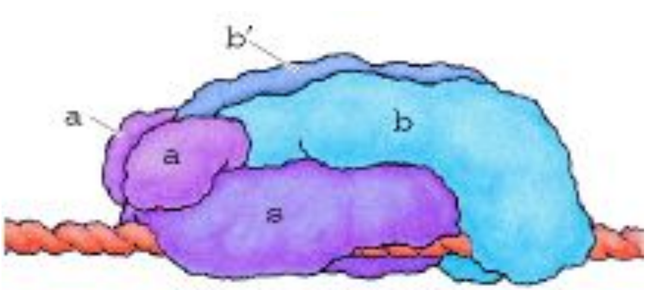
1. DNA unzips
2. mRNA is created using the DNA as a template
3. mRNA leaves the nucleus
4. REMEMBER!!!
   - A = U
   - G C

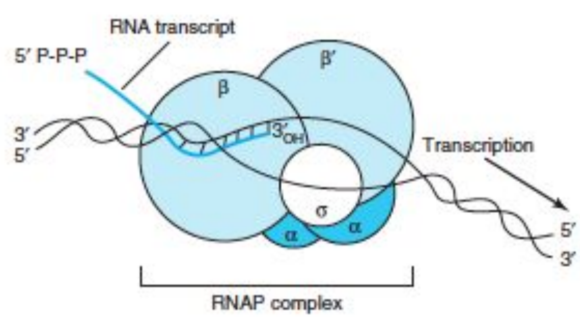
Transcription

- RNA Polymerase - a protein complex;
  - Reads a DNA template & synthesizing RNA,
  - Accessory proteins are also needed.
- Bacteria have one RNA polymerase, and eukaryotes have three (Pol-I, II & III).
- Prokaryotic and eukaryotic genes have promoter regions;
  - They point RNA polymerase in the right direction and position it to start.

Prokaryotic RNA polymerase

- Prokaryotic RNA pol.:
  - 4 core subunits
  - Sigma factor (σ)- determines promoter specificity
  - Core + σ = holoenzyme
  - Binds promoter sequence
  - Catalyzes “open complex” and transcription of DNA to RNA





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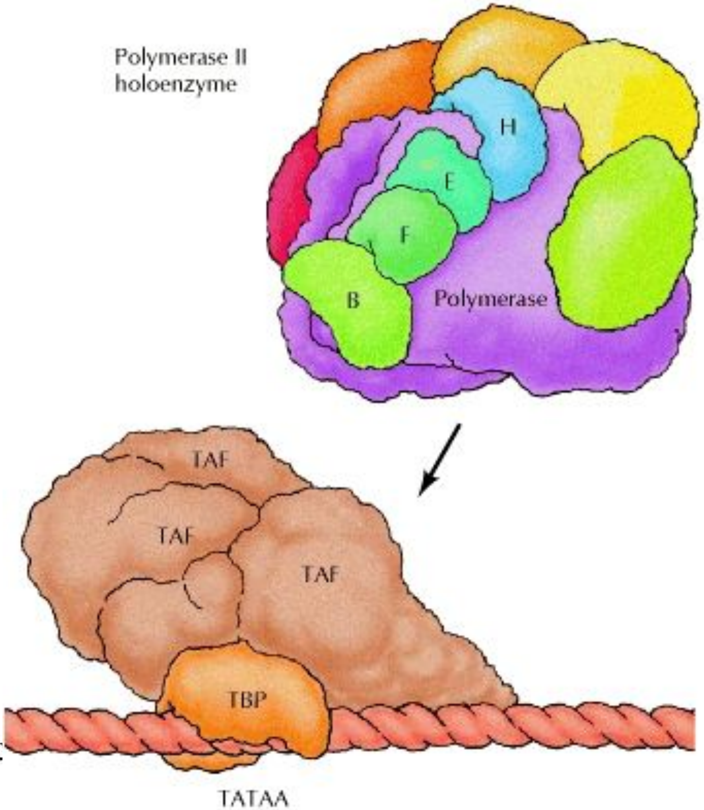
Eukaryotic RNA polymerases

Classes of Genes Transcribed by Eukaryotic RNA Polymerases

Type of RNA synthesized	RNA polymerase
Nuclear genes	
mRNA	II
tRNA	III
rRNA	
5.8S, 18S, 28S	I
5S	III
snRNA and scRNA	II and III[a]
Mitochondrial genes	Mitochondrial[b]
Chloroplast genes	Chloroplast[b]



a Some small nuclear (sn) and small cytoplasmic (sc) RNAs are transc e III.

b The mitochondrial and chloroplast RNA polymerases are similar to bacterial enzymes.

Pol I – has 14 subunits
Pol II – 10 -12 subunits
Pol III – 12 subunits

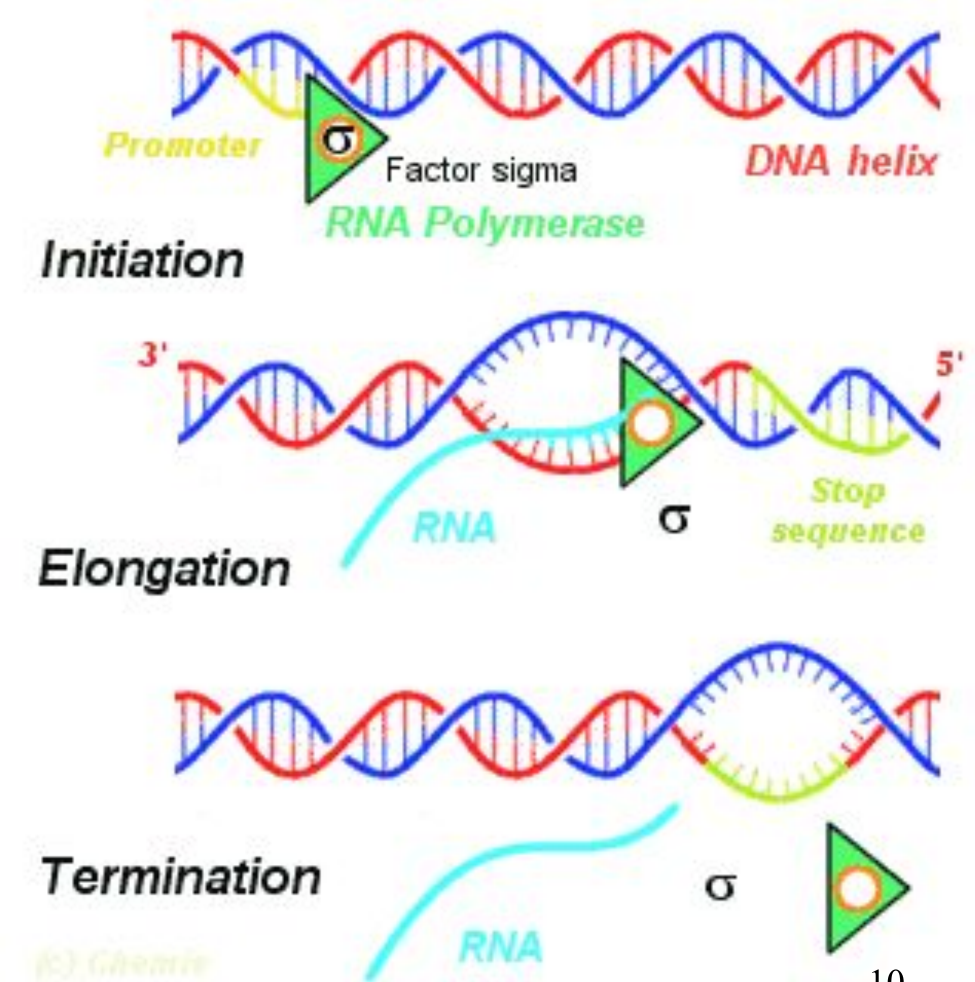
Comparison of RNA Polymerases

Prokaryotic	Eukaryotic
Single RNA polymerase ($\alpha_2 \beta\beta'$)	RNAP 1: rRNA (nucleolus) Except 5S rRNA RNAP 2: hnRNA/mRNA and some snRNA RNAP 3: tRNA, 5S rRNA
Requires sigma (σ) to initiate at a promoter	No sigma, but transcription factors (TFIID) bind before RNA polymerase
Sometimes requires rho (ρ) to terminate	No rho required
Inhibited by rifampin Actinomycin D	RNAP 2 inhibited by α-amanitin (mushrooms) Actinomycin D

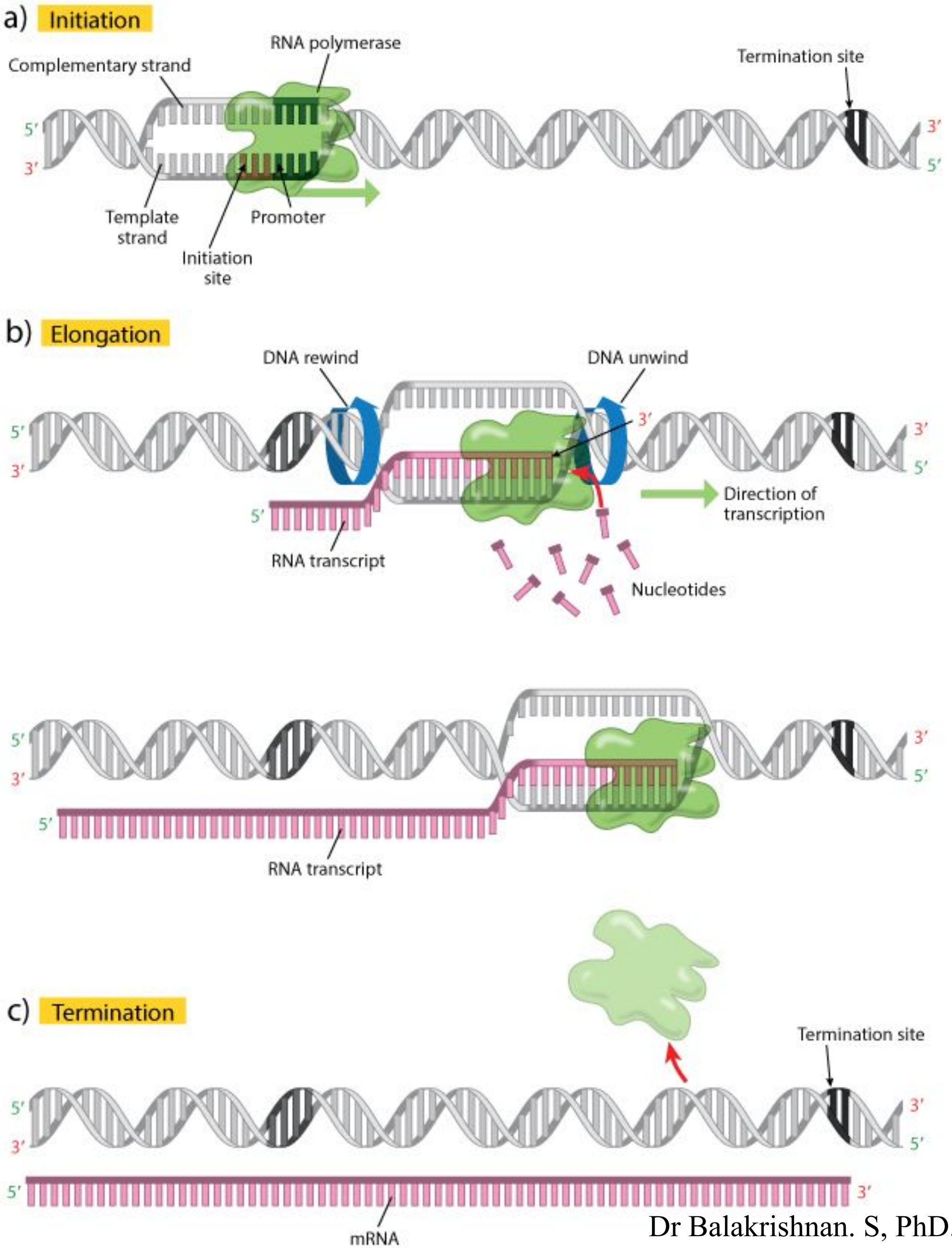
Phases of Transcription

Transcription has three major phases:

- Initiation
- Elongation
- Termination



Phases of Transcription



(A) The transcription process is initiated when the enzyme RNA polymerase binds to a DNA template at a promoter sequence.

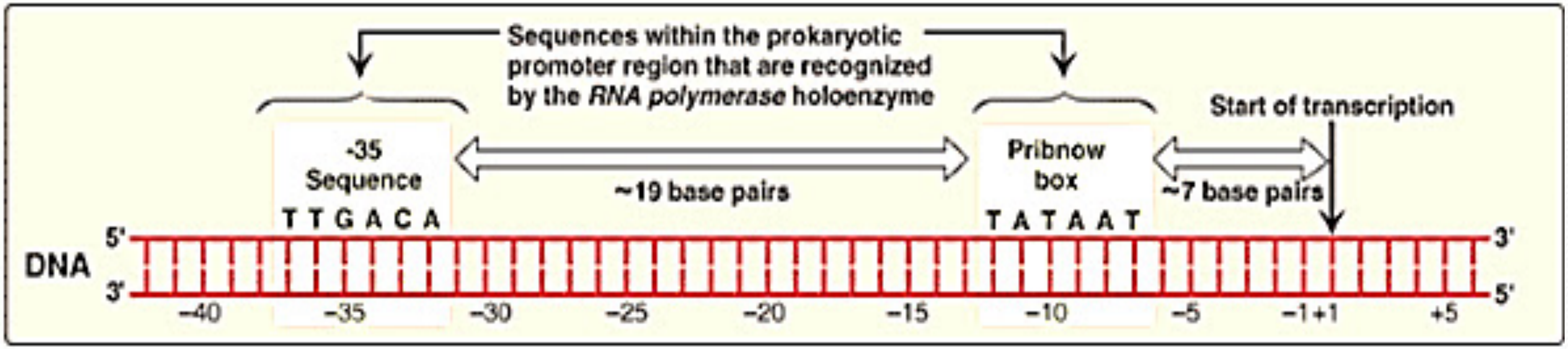
(B) During the elongation process, the DNA double helix unwinds. RNA polymerase reads the template DNA strand and adds nucleotides to the three-prime (3') end of a growing RNA transcript.

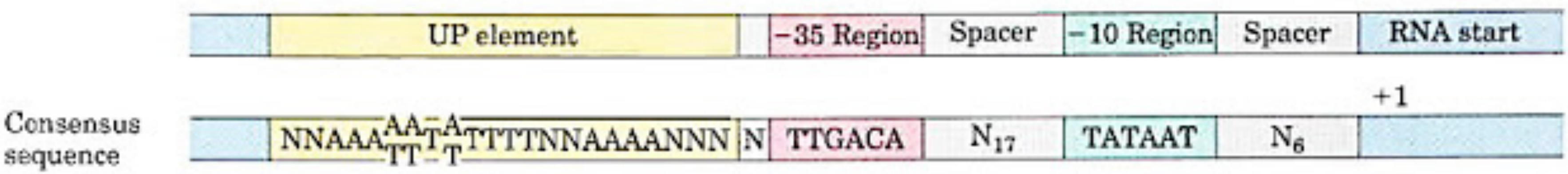
(C) When RNA polymerase reaches a termination sequence on the DNA template strand, transcription is terminated and the mRNA transcript and RNA polymerase are released from the complex.

Initiation

- RNA pol and accessory proteins bind to a DNA molecule upstream of the initiation point
- DNA unwound to separate expose the strand to be transcribed.
- Then, the RNA pol complex binds to a promoter sequence initiation of transcription.
- Pol begins to synthesize a strand of RNA complementary to one side of the DNA strand.
   - Purine is the 1st nucleotide to be polymerized and exists at 5' end of RNA as nucleotide.

Transcription Initiation



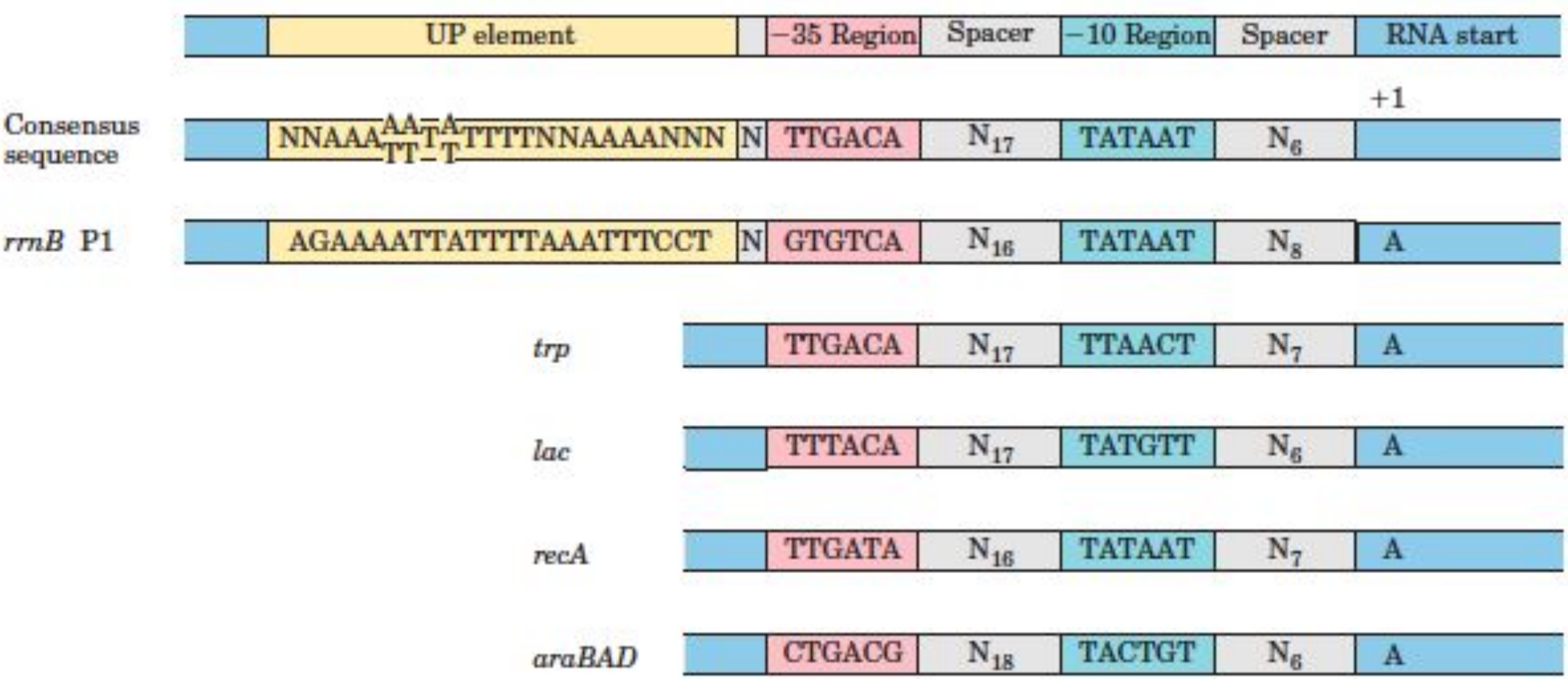


- Begins with the binding of RNA Pol holoenzyme to promoter in template DNA.

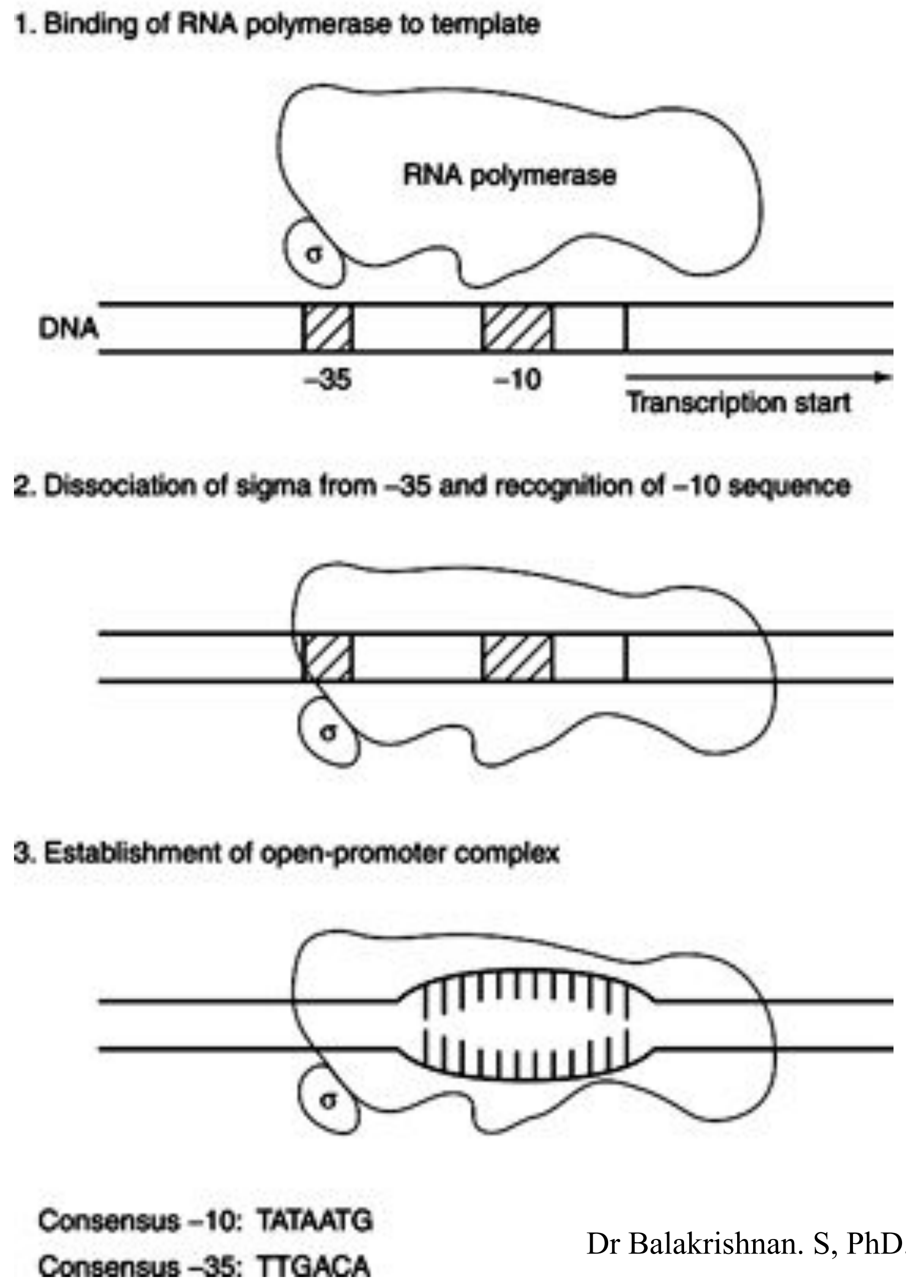
- Alpha subunit binds to UP element and sigma factor covers -35 to -10 region.

- Binding of RNAP to -35 forms closed complex whereas to -10 region (**Pribnow box or Pribnow-Schaller box**) forms open complex, due to the presence of AT region which has low nucleosomal density.

Promoters of Prokaryotes



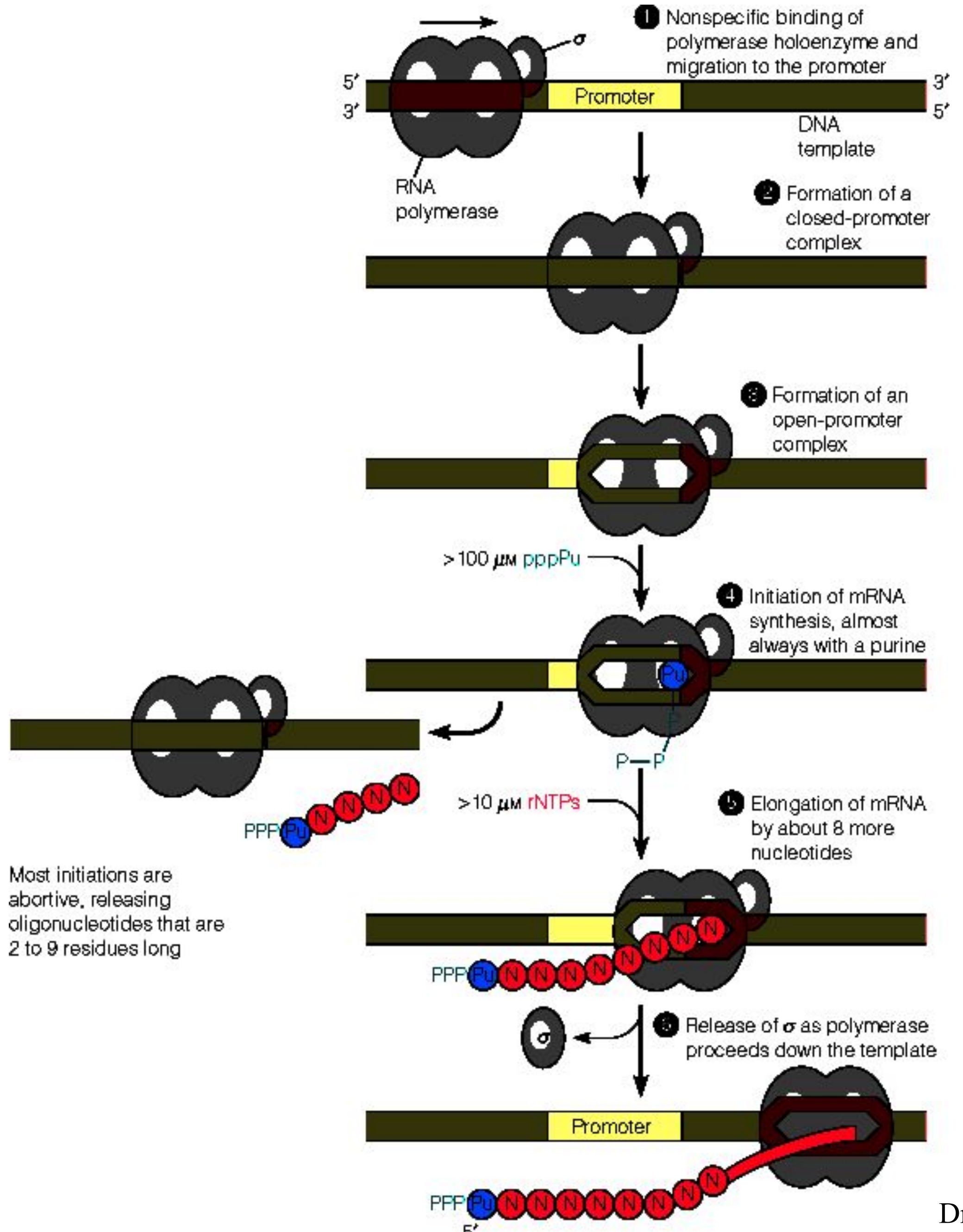
Transcription Initiation



Transcription Initiation

- The closed complex causes unwinding and separation of DNA strands to form **open (binary) promoter complex,**
  - After unwinding, only one of the two strands is copied; this is achieved by incorporation of nucleotides.
  - Initially without movement of enzyme leading to **the** formation of RNA chain, upto 9 bases in length.
  - During the incorporation of these 9 bases, at every **step,** there is a possibility for the release of this small RNA chain, a process described as **'abortive initiation'.**
  - A cycle of abortive initiations usually 'occurs generating a series of short (2-9 base) oligonucleotides, before initiation is actually Successful.
  - Once the transcript reaches approximately 23 nucleotides, it no longer slips and elongation can occur.
- Once initiation succeeds, the sigma factor of RNA polymerase dissociates.
- The dissociation of σ factor marks the entry of NusA protein, which helps elongation, and promotes pausing and termination at specific sites.

Transcription Initiation



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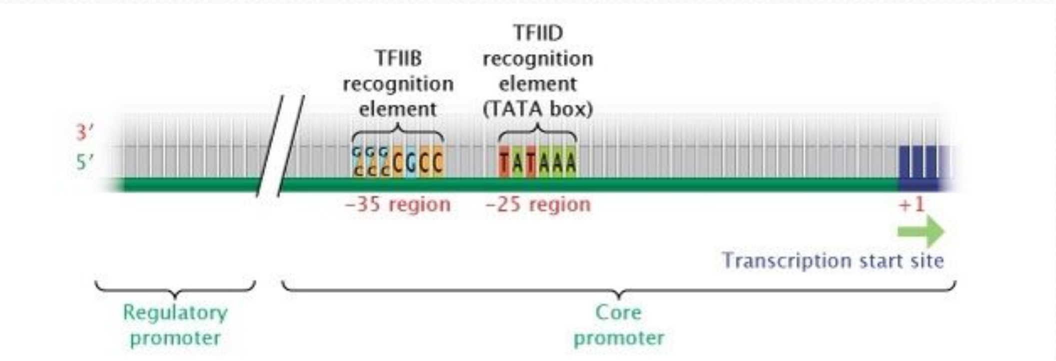
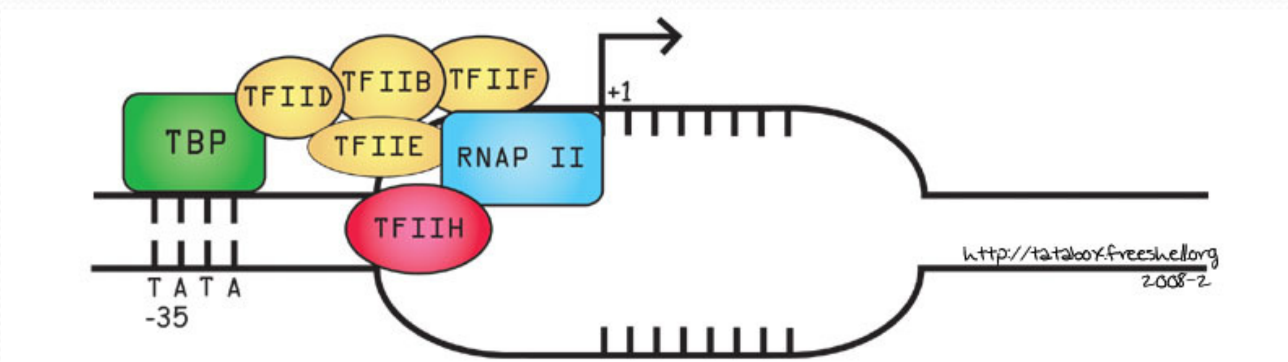
Promoters of Eukaryotes

- Promoters - **TATA box** (-25 to -35, Goldberg-Hogness box), **CCAAT box** (-70, sometimes called as CAAT or CAT box) & **GC box** (-110, act as enhancer)
- RNA Polymerase II, Transcription factors – TFIIA, TFIIB,TFIIF, TFIIE, TFIIH (two activity - unwinding of DNA & Phosphorylates RNA Pol II)), TATA binding protein (TBP) & TBP associated factors (TAFs; also called TFIID) load & forms transcription initiation complex DNA unwinds transcription begins



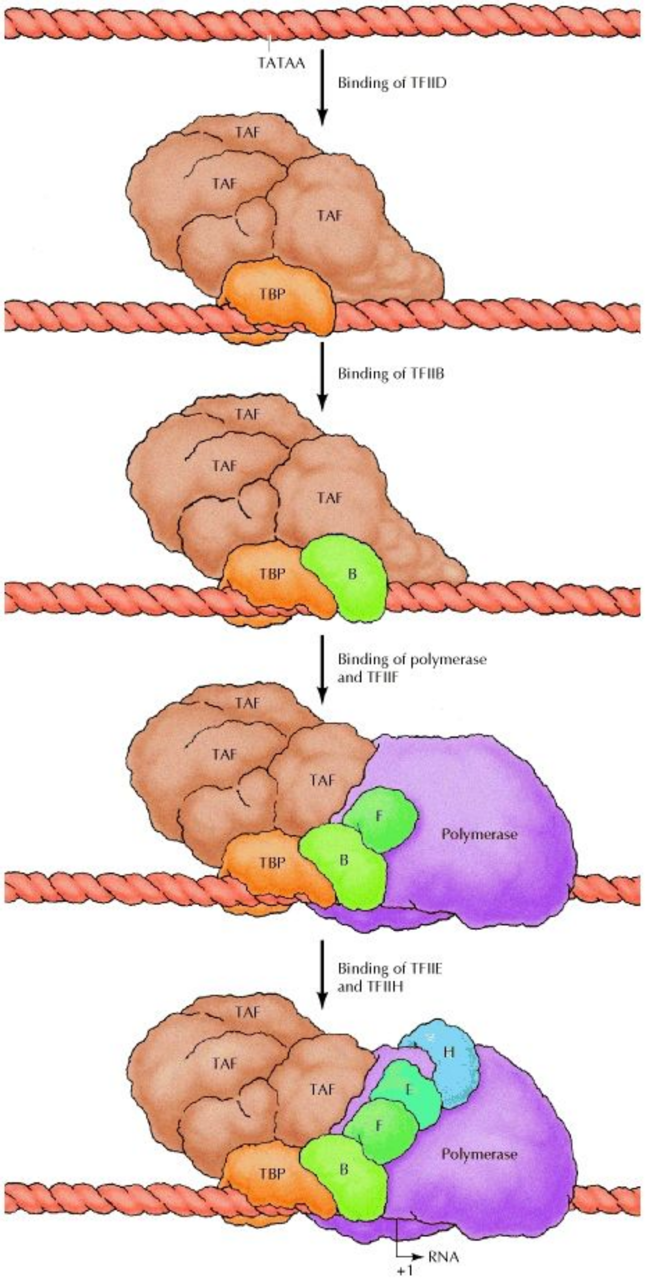


Initiation in Eukaryotes

TABLE 26–1 Proteins Required for Initiation of Transcription at the RNA Polymerase II (Pol II) Promoters of Eukaryotes

Transcription protein	*Number of subunits*	*Subunit(s)* M_r	*Function(s)*
Initiation			
Pol II	12	10,000–220,000	Catalyzes RNA synthesis
TBP (TATA-binding protein)	1	38,000	Specifically recognizes the TATA box
TFIIA	3	12,000, 19,000, 35,000	Stabilizes binding of TFIIB and TBP to the promoter
TFIIB	1	35,000	Binds to TBP; recruits Pol II–TFIIF complex
TFIIE	2	34,000, 57,000	Recruits TFIIH; has ATPase and helicase activities
TFIIF	2	30,000, 74,000	Binds tightly to Pol II; binds to TFIIB and prevents binding of Pol II to nonspecific DNA sequences
TFIIH	12	35,000–89,000	Unwinds DNA at promoter (helicase activity); phosphorylates Pol II (within the CTD); recruits nucleotide-excision repair proteins

Formation of a polymerase II transcription complex



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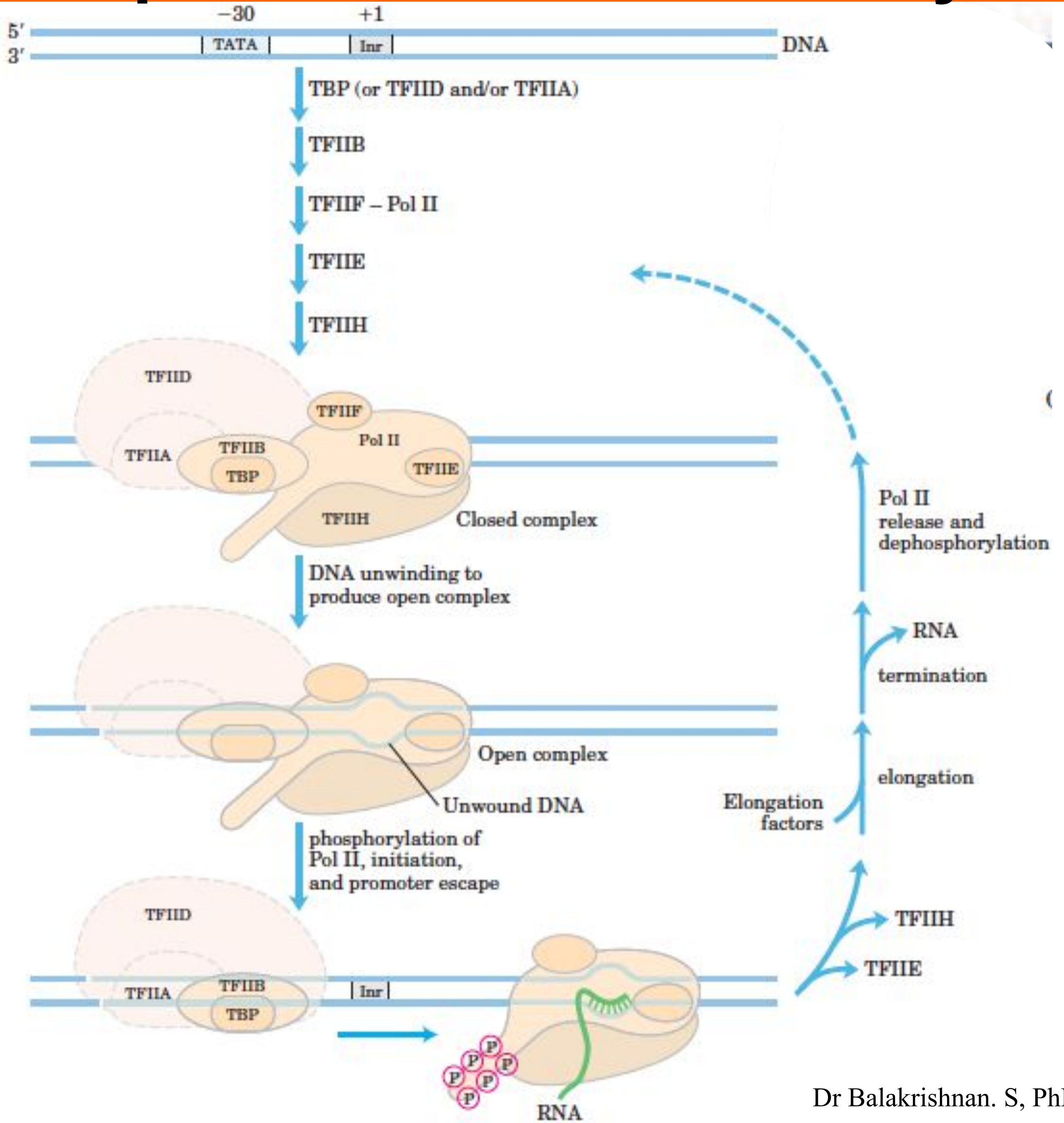
Transcription Initiation in Eukaryotes



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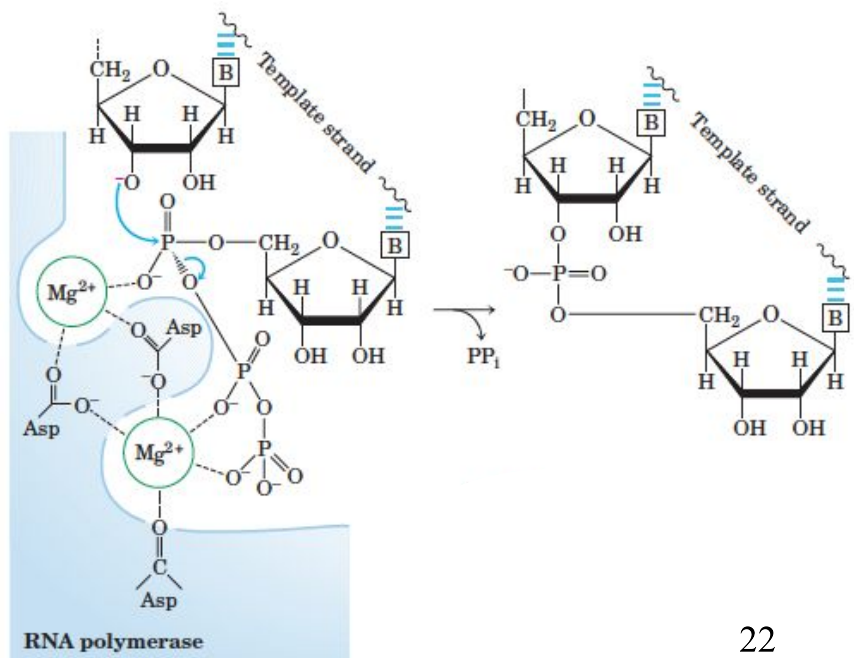
Elongation

- During elongation, a lengthening RNA molecule is produced by RNA polymerase as it reads the DNA triplet code on the template strand.
  - RNA polymerase elongates an RNA strand by adding ribonucleotide units to the 3 -hydroxyl end, building RNA in the 5’ → 3’ direction.
  - The 3’-hydroxyl group acts as a nucleophile, attacking the α-phosphate of the incoming ribonucleoside triphosphate and releasing pyrophosphate.
- The overall reaction is:

$$\underset{\text{RNA}}{(\text{NMP})_n} + \text{NTP} \longrightarrow \underset{\text{Lengthened RNA}}{(\text{NMP})_{n+1}} + \text{PP}_i$$



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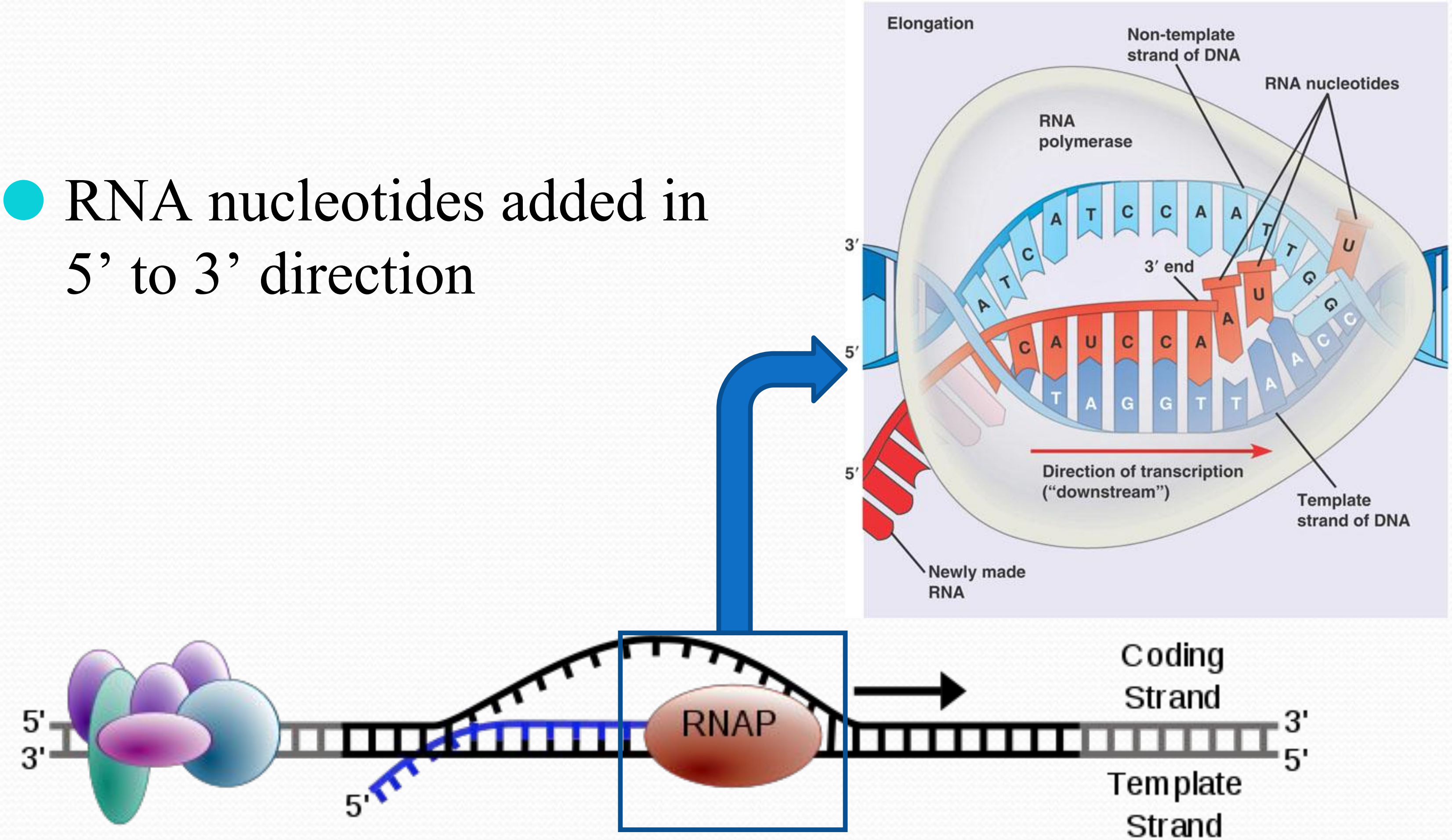
Elongation

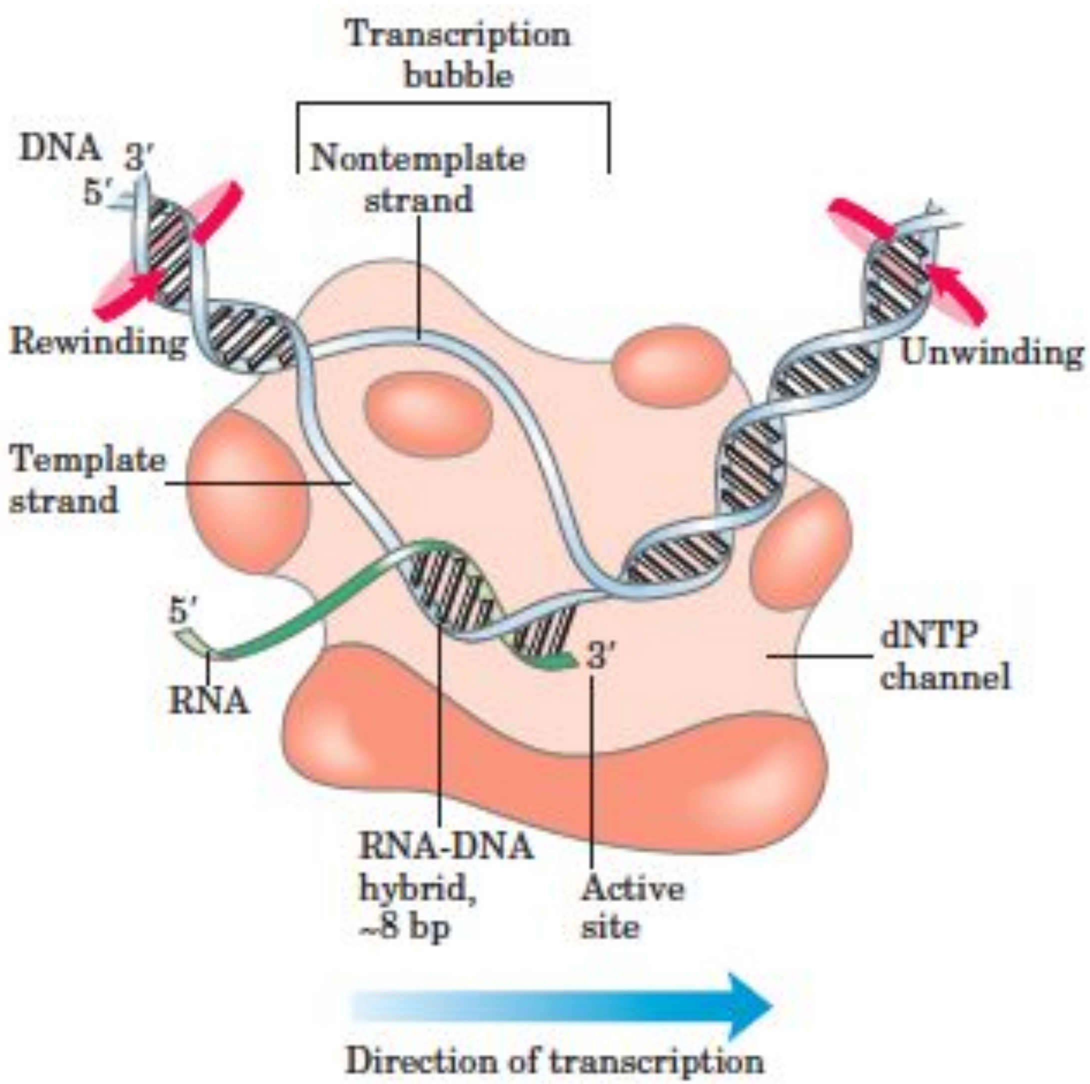
- The RNA polymerase core enzyme will continue reading the template until it reaches a end signal.
    - It moves along the template, locally "unzipping" the DNA double helix.
    - This allows a transient base pairing between the incoming nucleotide and newly-synthesized RNA and the DNA template strand.
    - As it is made, the RNA transcript forms secondary structure through intra-strand base pairing.
- The average speed of transcription is about 40 nucleotides per second, much slower than DNA polymerase.
- Eventually, RNA polymerase must come to the end of the region to be transcribed;
    - Another RNA polymerase can attach to the promoter to begin synthesizing another RNA before the first one is finished.

Transcription: Elongation

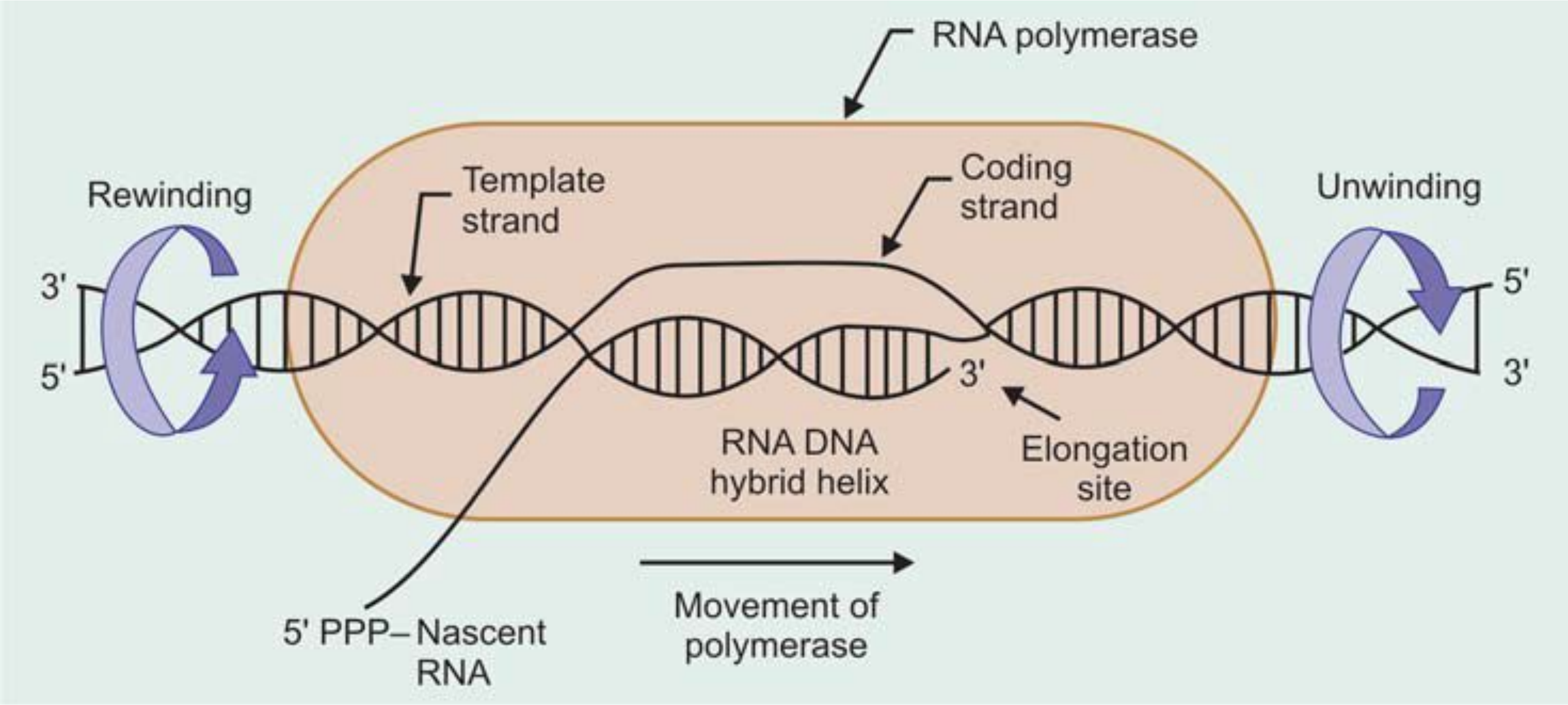
- RNA nucleotides added in 5’ to 3’ direction



Elongation: Transcription bubble



Elongation: Transcription bubble



Termination

- A signal that tells RNA polymerase to stop, or terminate, transcription.
- RNA polymerase loses affinity for the DNA template when it encounters a particular DNA sequence.
  - At this point, it disengages from the DNA
  - The RNA molecule is released for translation or post-transcriptional processing.
- In prokaryotes, this signal can take two forms:
  - rho-independent
  - rho-dependent.
- In eukaryotes – not completely known;
  - May be assisted by ‘poly A’ signal
  - Termination signal on the DNA

Rho-independent Termination

- More simple - called simple termination.

- A signal is found on the DNA template - consists of a region that contains a section that is then repeated a few base pairs away in the inverted sequence (palindromic sequence).

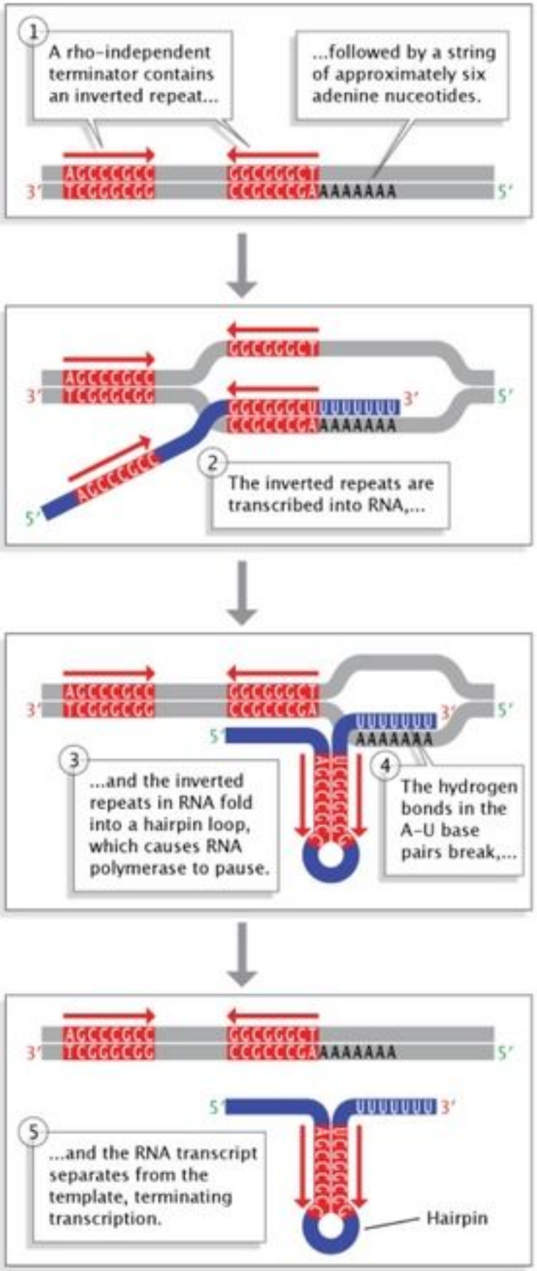
    - The RNA can fold back and base pair with itself forming a hairpin loop.

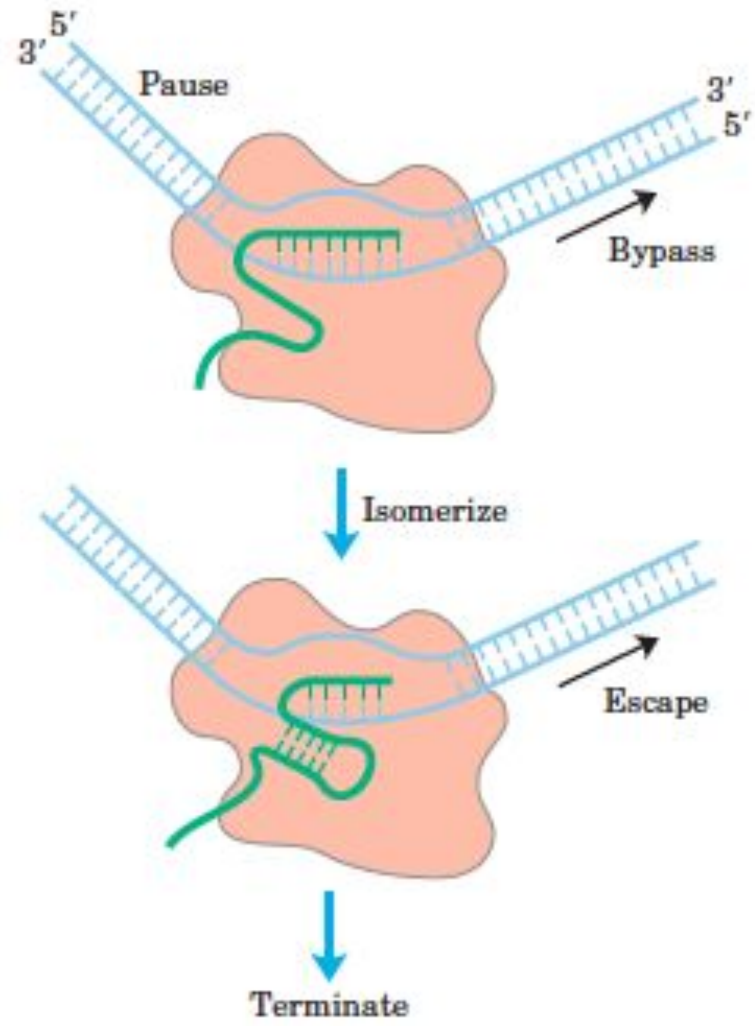
- Then, a string of adenines in the DNA sequence are transcribed into uracils in the RNA sequence.

    - The uracil bases will only pair weakly with the adenines.

    - So, the RNA chain can easily be released from the DNA template, terminating transcription.

Rho-independent Termination





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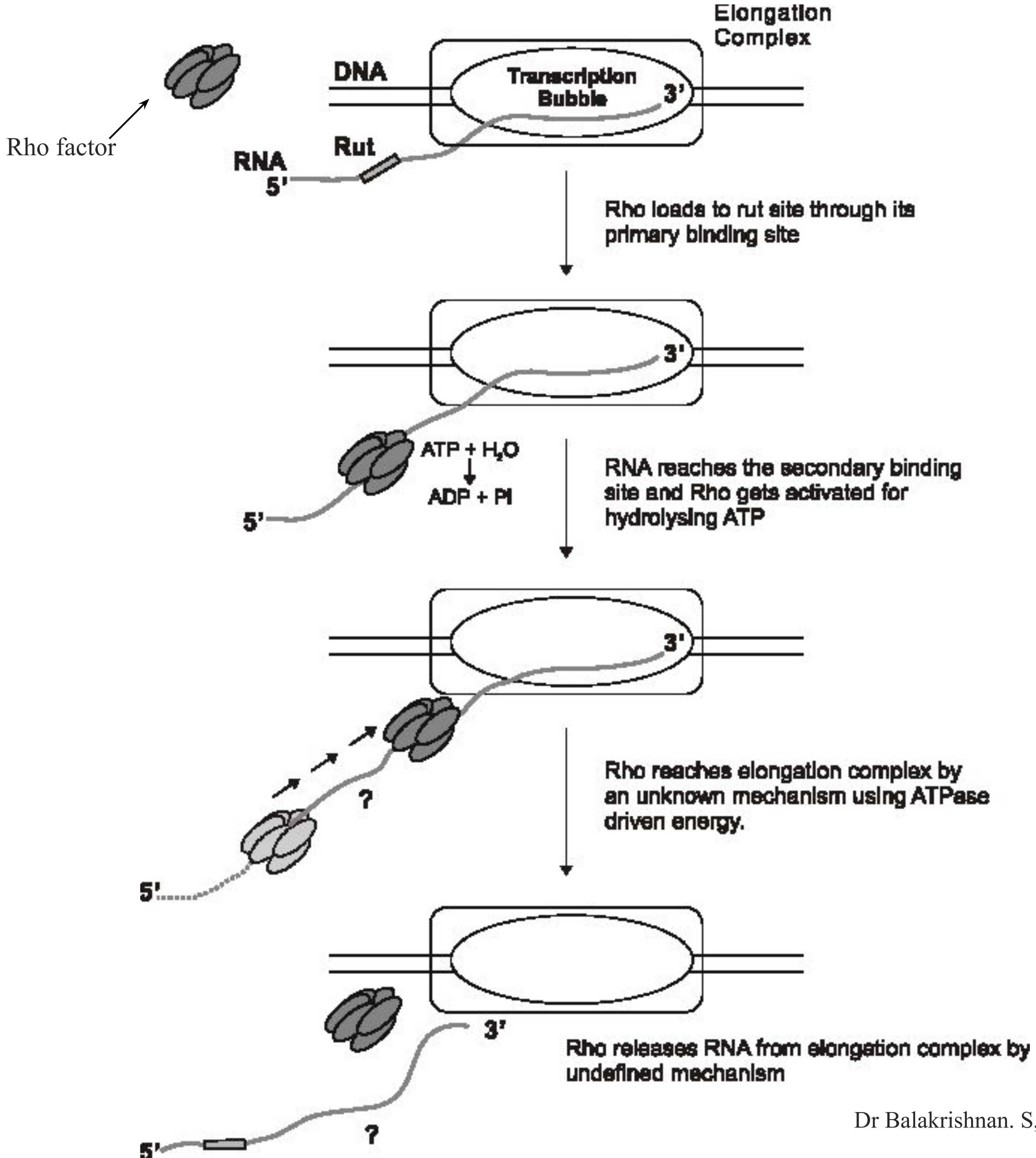
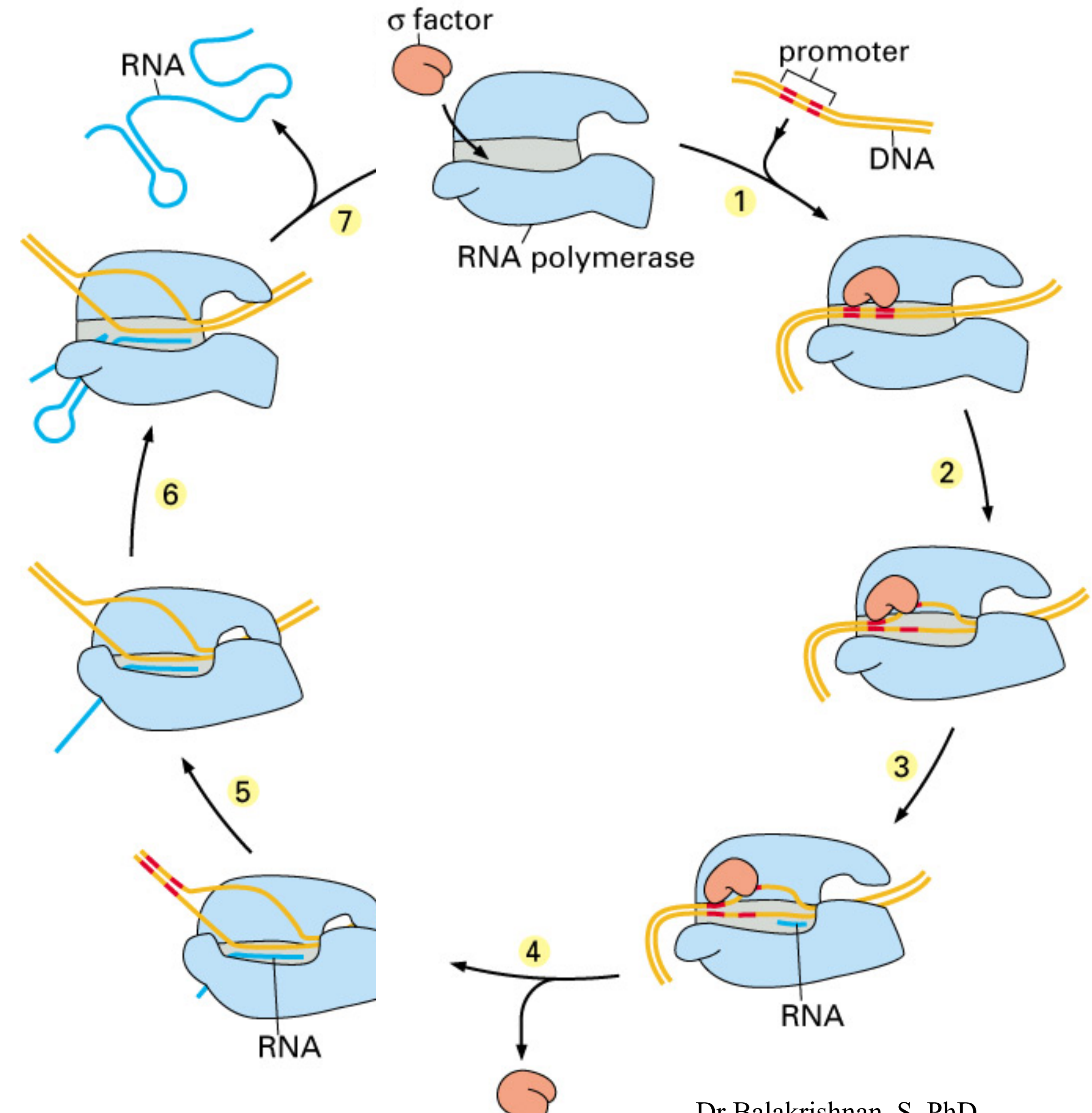
Rho-Dependent Termination

- Dependent on a specific protein called a rho factor.
   - Bind to the end of the RNA chain and slide along the strand towards the open complex bubble.
   - When it catches the polymerase, causes the termination of transcription.
- The mechanism of this termination is unclear;
   - But the rho factor could in some way pull the polymerase complex off of the DNA strand.

Rho-Dependent Termination



Prokaryotic Transcription: Summary



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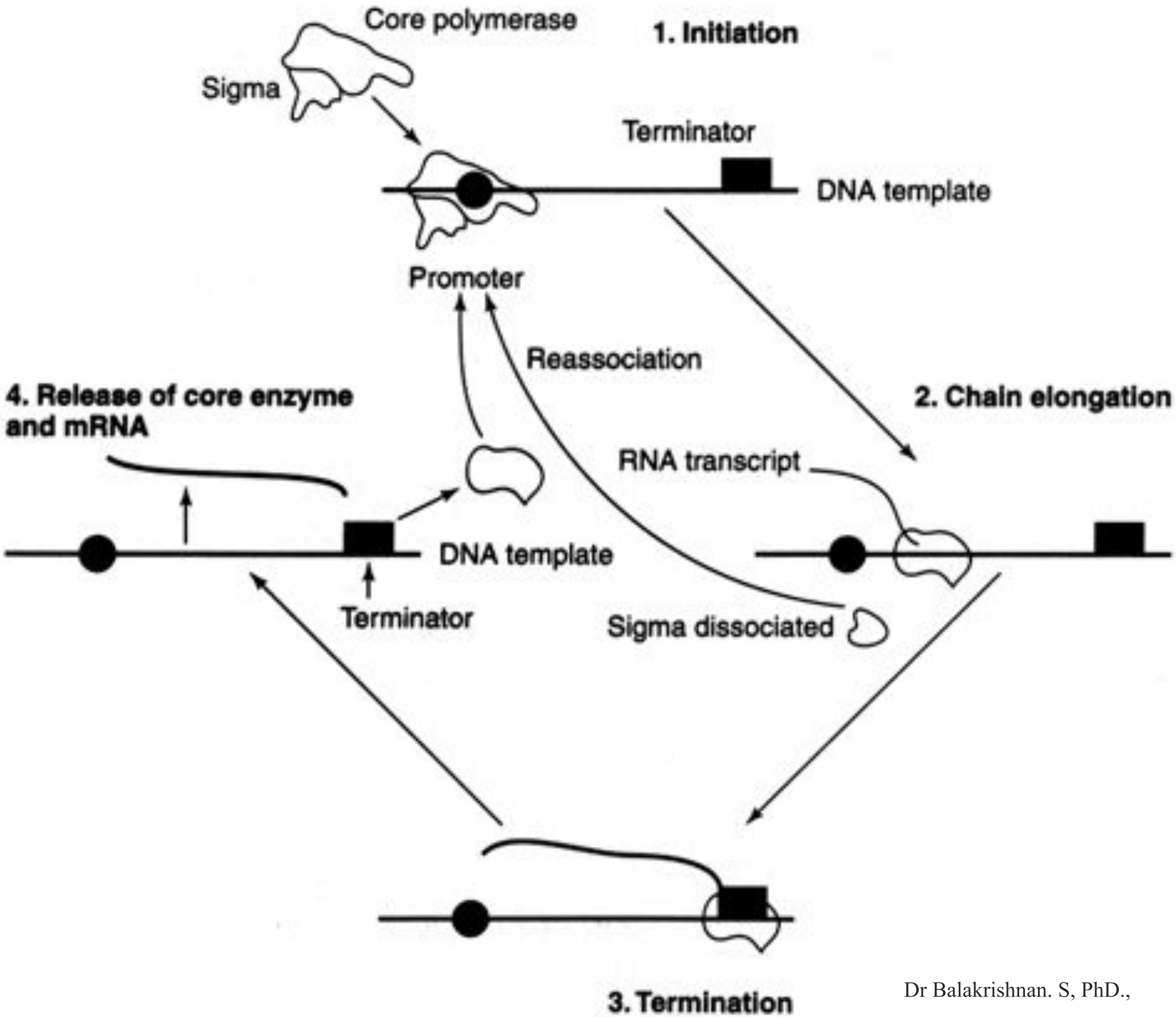
Transcription: summary



Primary Transcript

- Primary Transcript-the initial molecule of RNA produced--- hnRNA (heterogenous nuclear RNA)

- In prokaryotes, DNA →RNA →protein in cytoplasm concurrently

- In eukaryotes nuclear RNA >> Cp RNA

RNA processing: processes

A) Cutting and trimming to generate ends of;

- rRNA, tRNA and mRNA

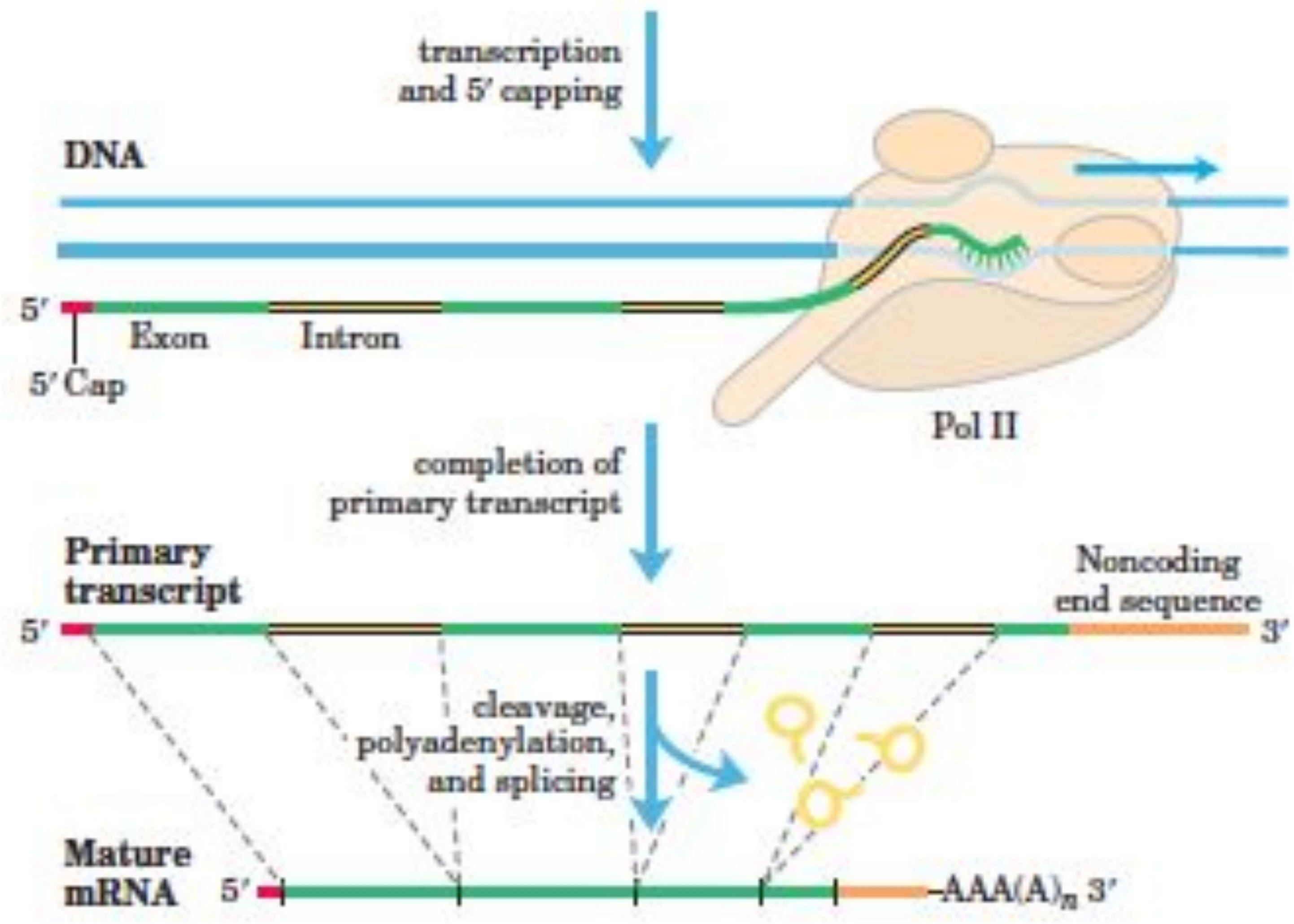
B) Covalent modification to add;

- A cap and a polyA tail to mRNA
- A methyl group to 2’-OH of ribose in mRNA and rRNA
- Extensive changes of bases in tRNA

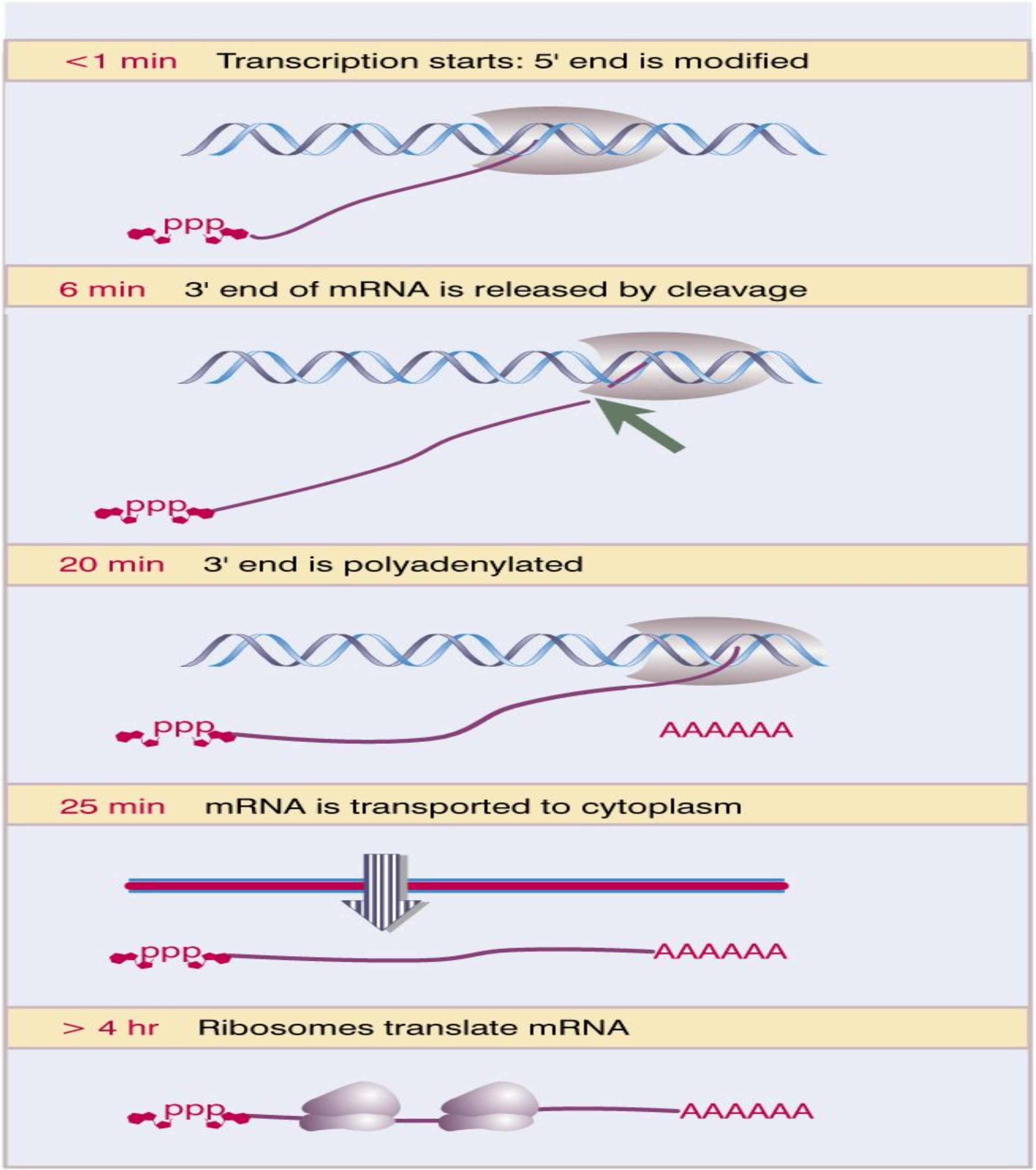
C) Splicing:

- pre-rRNA, pre-mRNA, pre-tRNA by different mechanisms.

RNA processing



Time course of RNA processing



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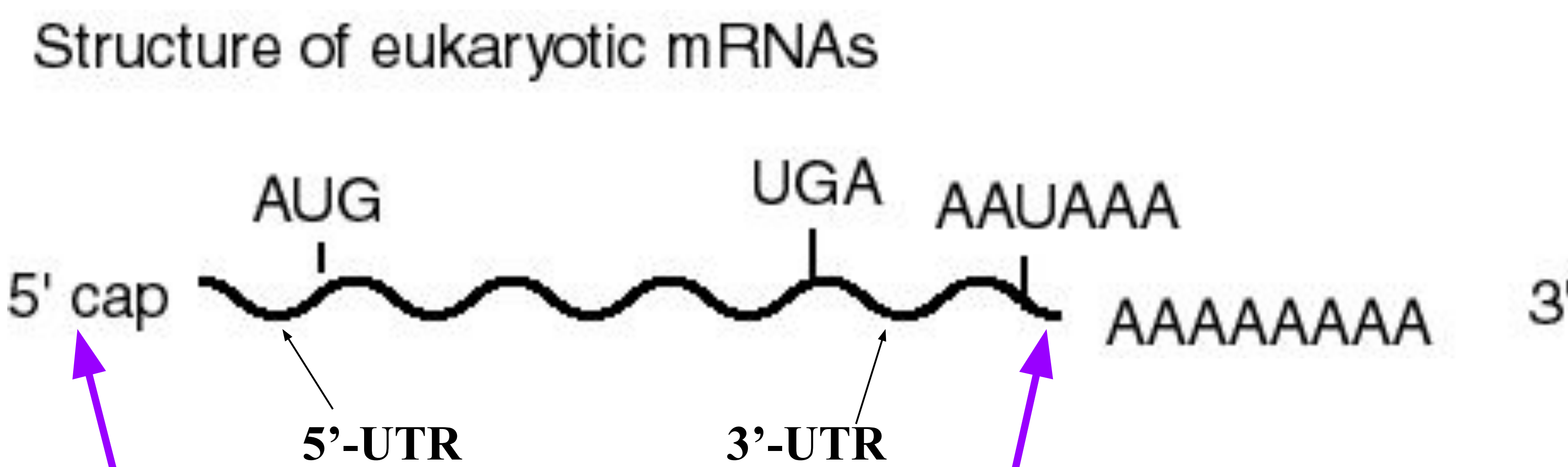
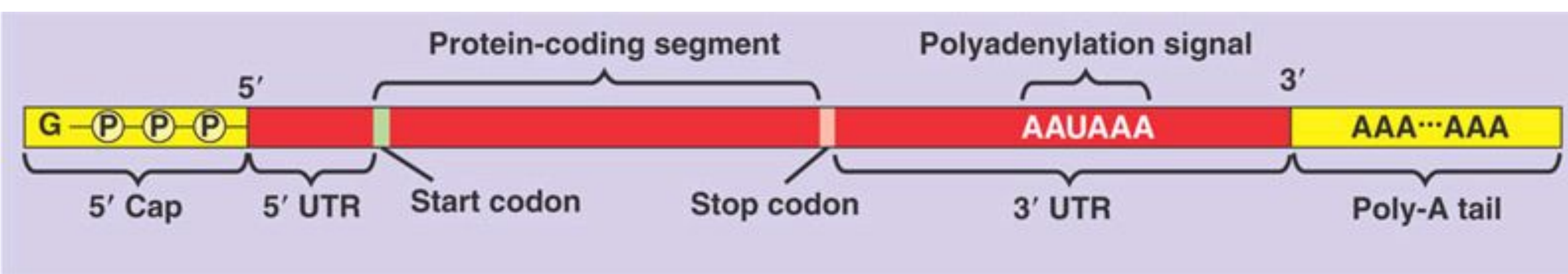
5’ and 3’ ends of eukaryotic mRNA



Add a GMP
Methylate it and
1st few nucleotides

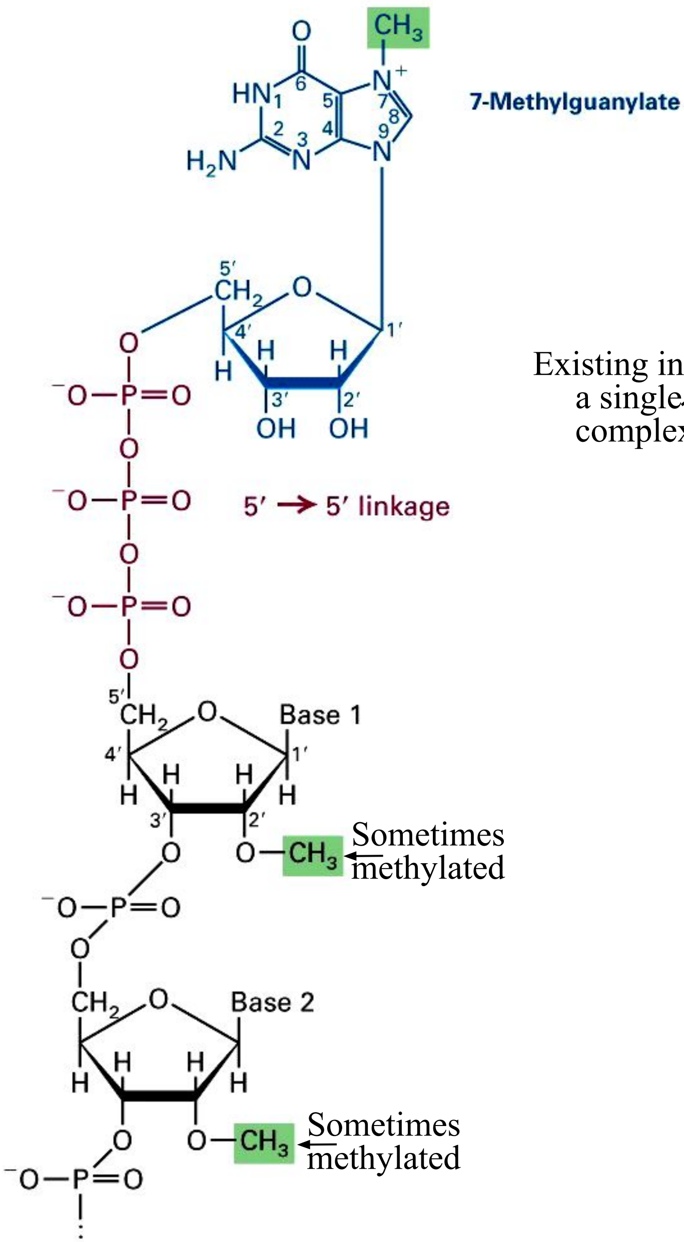
Cut the pre-mRNA
and add A’s



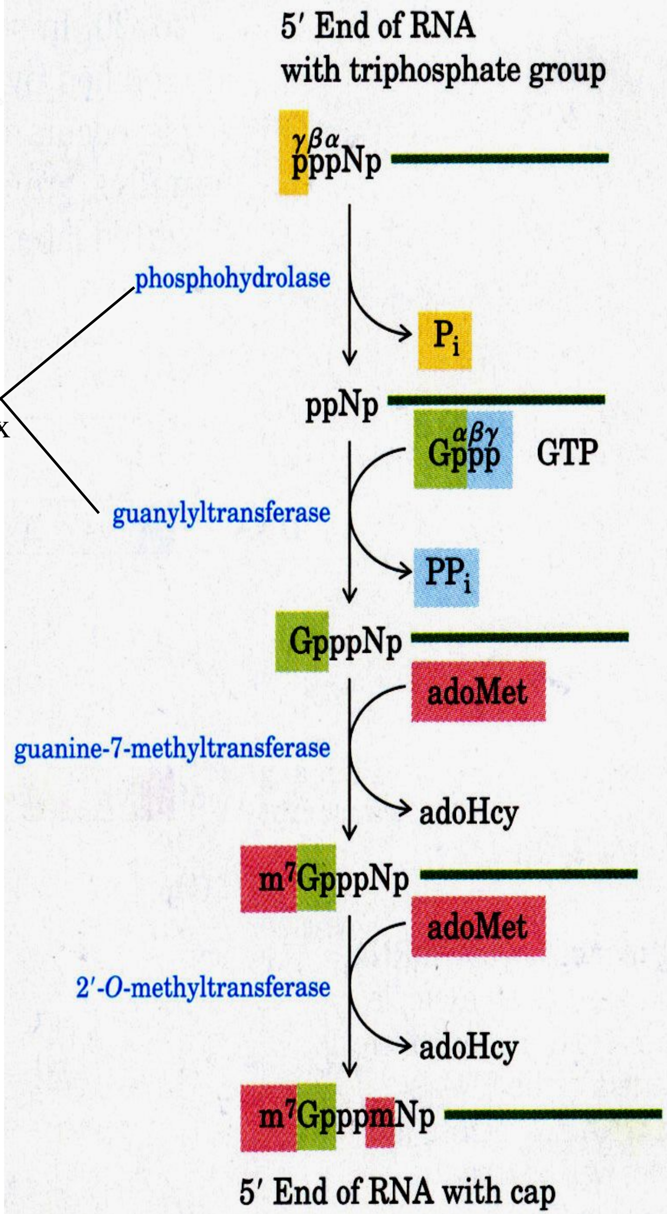
Capping of pre-mRNAs

- Cap=modified guanine nucleotide
- Capping= first mRNA processing event - occurs during transcription
- CTD (C-terminal repeat domain of Pol-II) recruits capping enzyme as soon as it is phosphorylated
- Pre-mRNA modified with 7-methyl-guanosine triphosphate (cap) when RNA is only 25-30 bp long
- Cap structure is recognized by CBC（cap-binding complex). It;
    - Stablizes the transcript
    - Prevents degradation by exonucleases
    - Stimulates splicing and processing

Capping of the 5' end of nascent RNA transcripts with m^{7}G

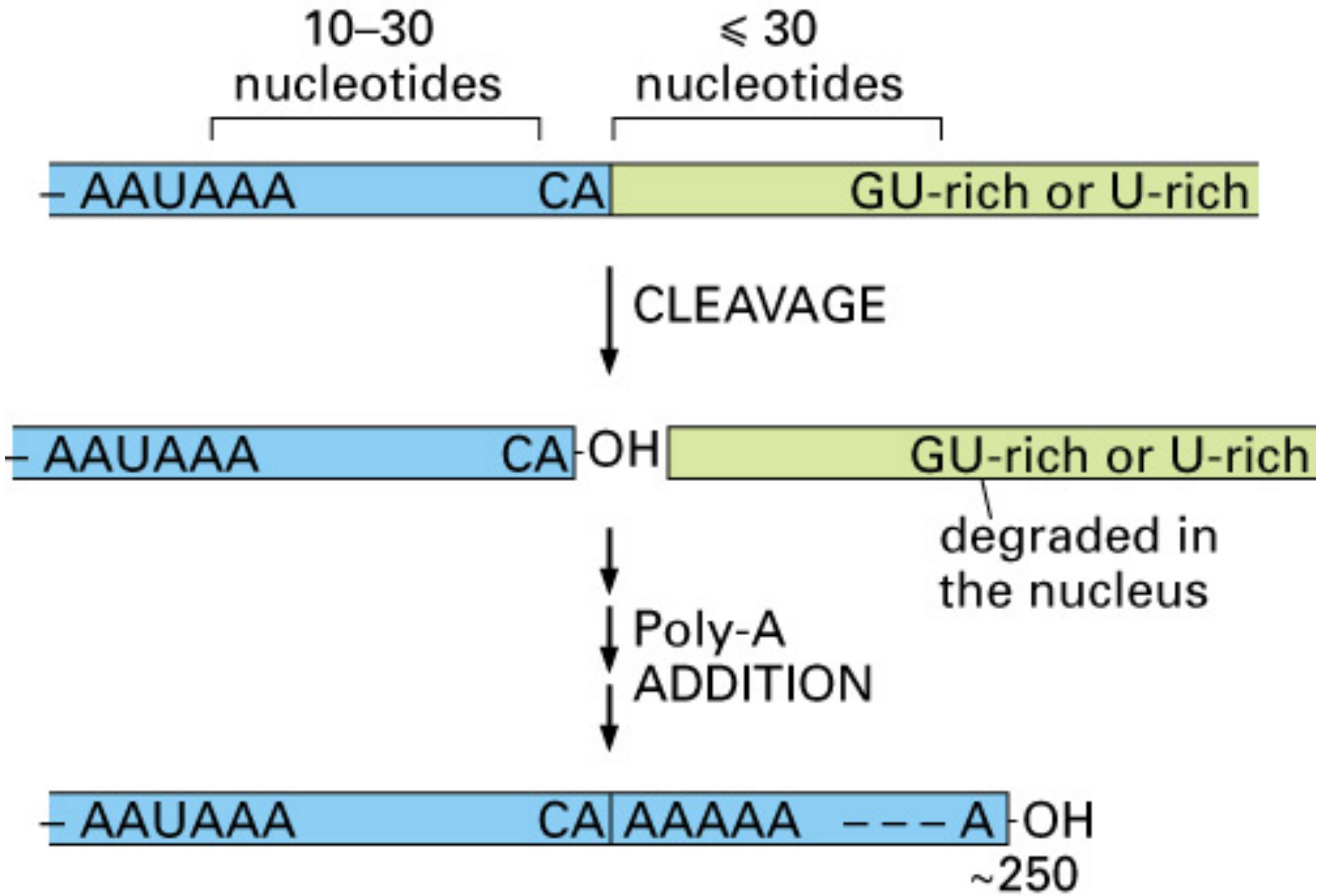


Existing in a single complex



- The cap is added after the nascent RNA molecules produced by RNA polymerase II reach a length of 25-30 nucleotides. Guanylyltransferase is recruited and activated through binding to the Ser5-phosphorylated Pol II CTD.
- The methyl groups are derived from S-adenosylmethionine.

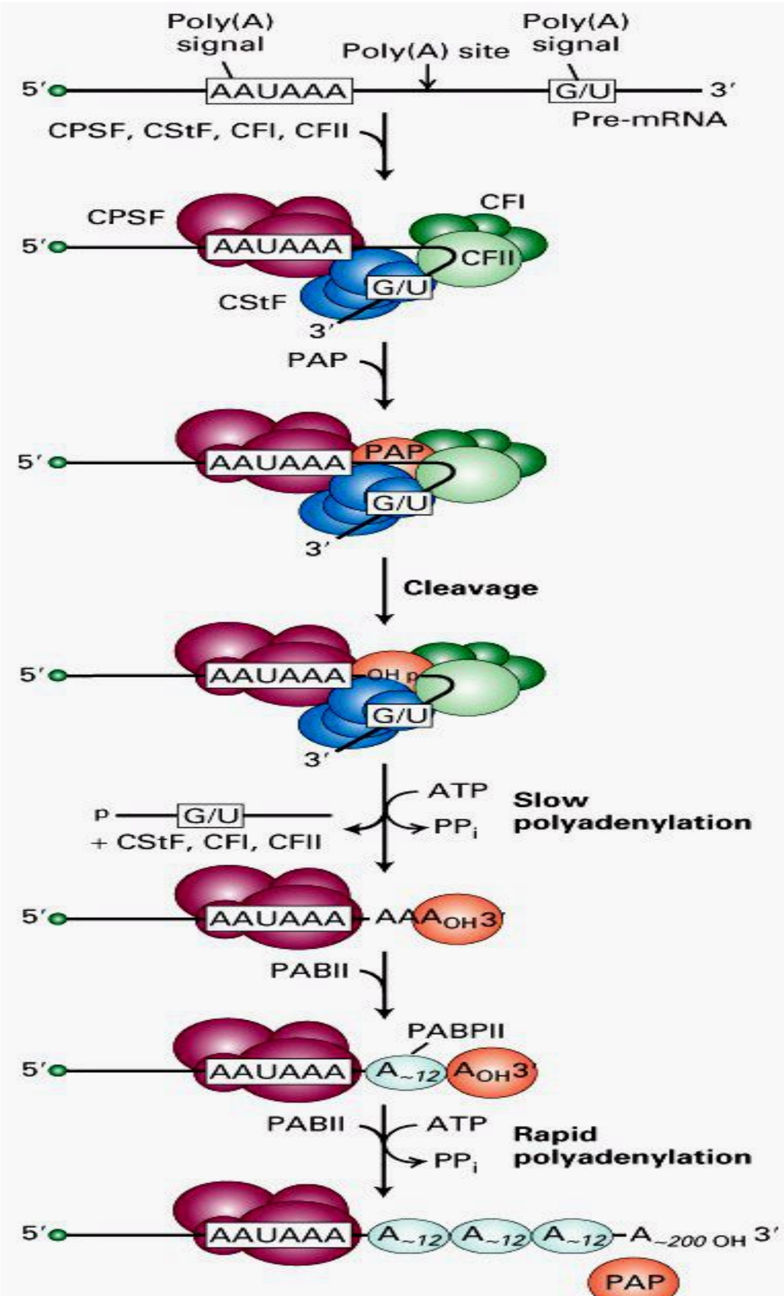
Consensus sequence for 3' process



AAUAAA: CstF (cleavage stimulation factor F)

GU-rich sequence: CPSF (cleavage and polyadenylation specificity factor)

Polyadenylation of mRNA at the 3’ end



- CPSF: *cleavage and polyadenylation specificity factor.*
  - CStF: *cleavage stimulatory factor.*
  - CFI & CFII: *cleavage factor I & II.*
  - PAP: *poly(A) polymerase.*
  - PABPII: *poly(A)-binding protein II.*
- RNA is cleaved 10~35nts 3’ to AAUAAA.
- The binding of PAP prior to cleavage ensures that the free 3’ end generated is rapidly polyadenylated.
- PAP adds the first 12A residues to 3’-OH slowly.
- Binding of PABPII to the initial short poly(A) tail accelerates polyadenylation by PAP.

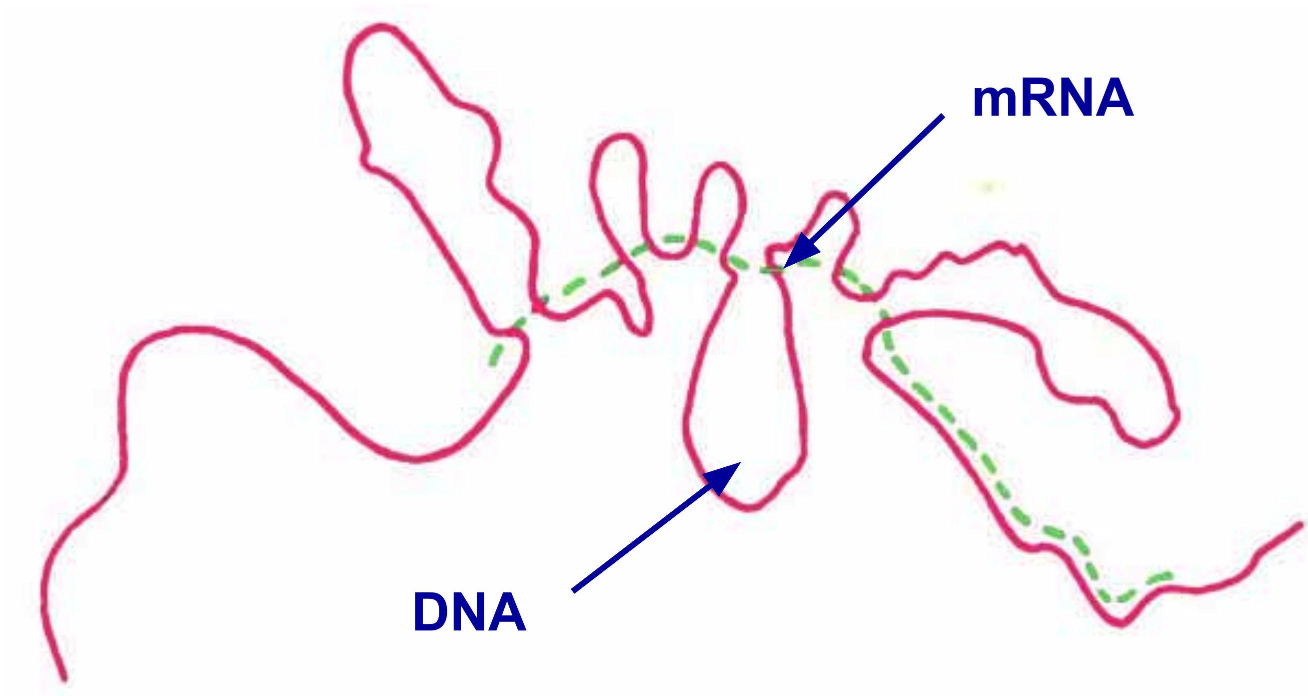
   Poly(A) tail stabilizes mRNA
   Enhances translation
   Help in export into the cytoplasm.

- The polyadenylation complex is associated with the CTD of Pol II following initiation.

Functions of 5' cap and 3' polyA

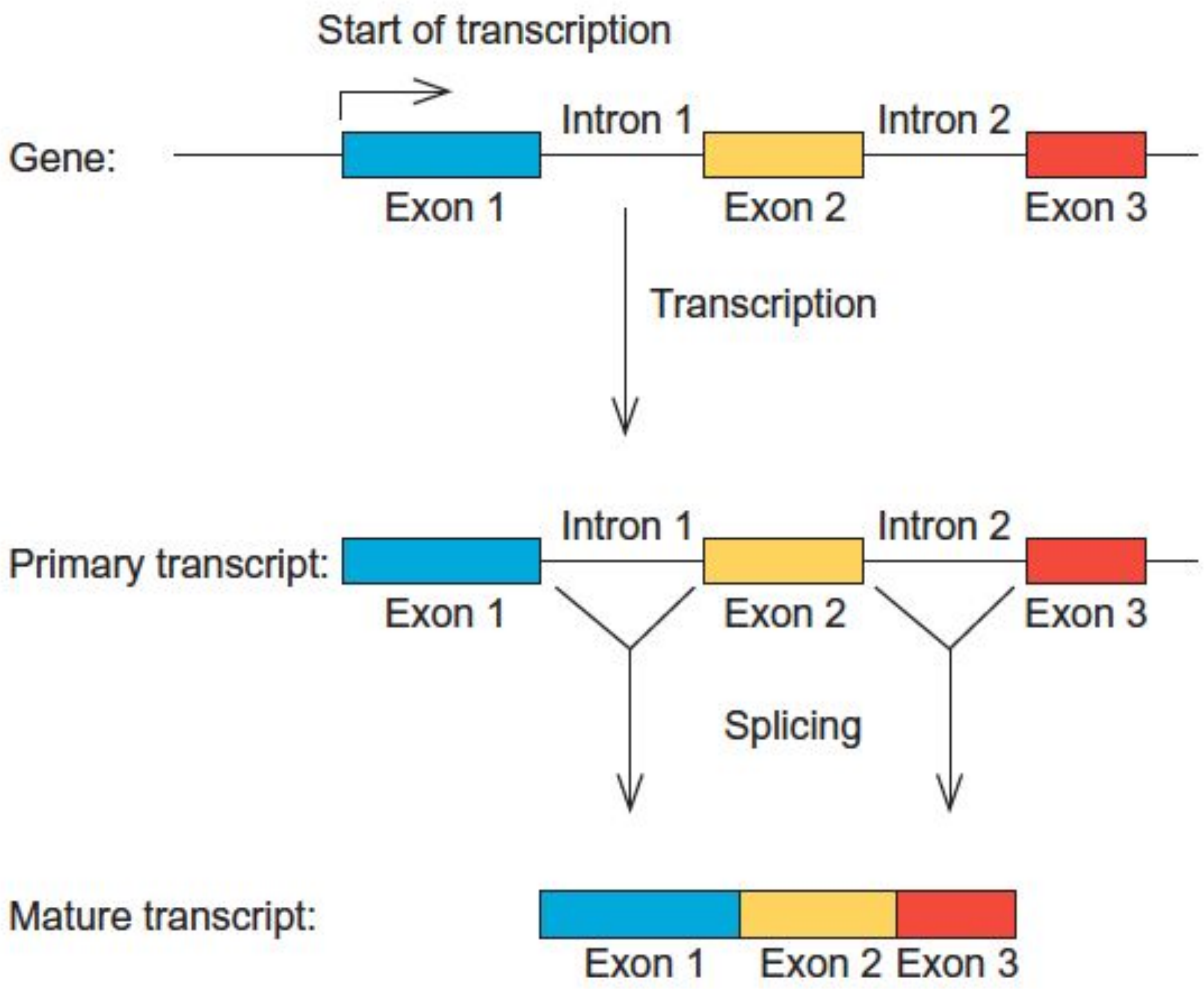
- Need 5' cap for efficient **translation**:
- Eukaryotic translation initiation factor 4 (eIF4) recognizes and binds to the cap as part of initiation.
- Both cap and polyA contribute to **stability** of mRNA:
- Most mRNAs without a cap or polyA are degraded rapidly.
- Shortening of the polyA tail and decapping are part of one pathway for RNA degradation in yeast.

Split gene and mRNA splicing



The matured mRNAs are much shorter than the DNA templates.

Outline of splicing



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Exon and Intron

- **Exon:**
  - Coding sequences
  - Any segment of an interrupted gene
  - Represented in the mature RNA product.
  - Are similar in size.
- **Intron:**
  - Intervening sequences
  - A segment of DNA that is transcribed, but removed from the transcript by splicing.
  - Are highly variable in size.

Types of introns

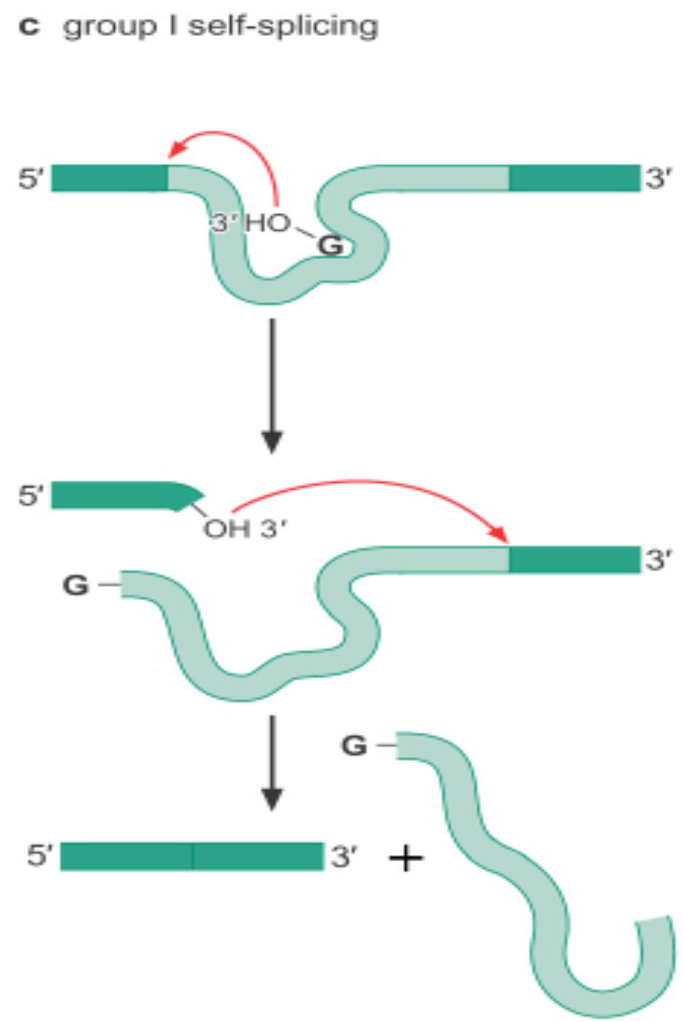
- 4 major types - On the basis of their mechanism of splicing and/or characteristic sequences.
  - Group I introns: In fungal mitochondria, plastids, and in pre-rRNA in Tetrahymena (self-splicing)
  - Group II introns: In fungal mitochondria and plastids (self-splicing)
  - Introns in pre-mRNA (spliceosome mediated)
  - Introns in pre-tRNA

Splicing of Group I introns

Group I

- **Self-splicing**

- Initiate splicing with a **G** nucleotide
- Uses a phosphoester transfer mechanism
- Does **not** require ATP hydrolysis.

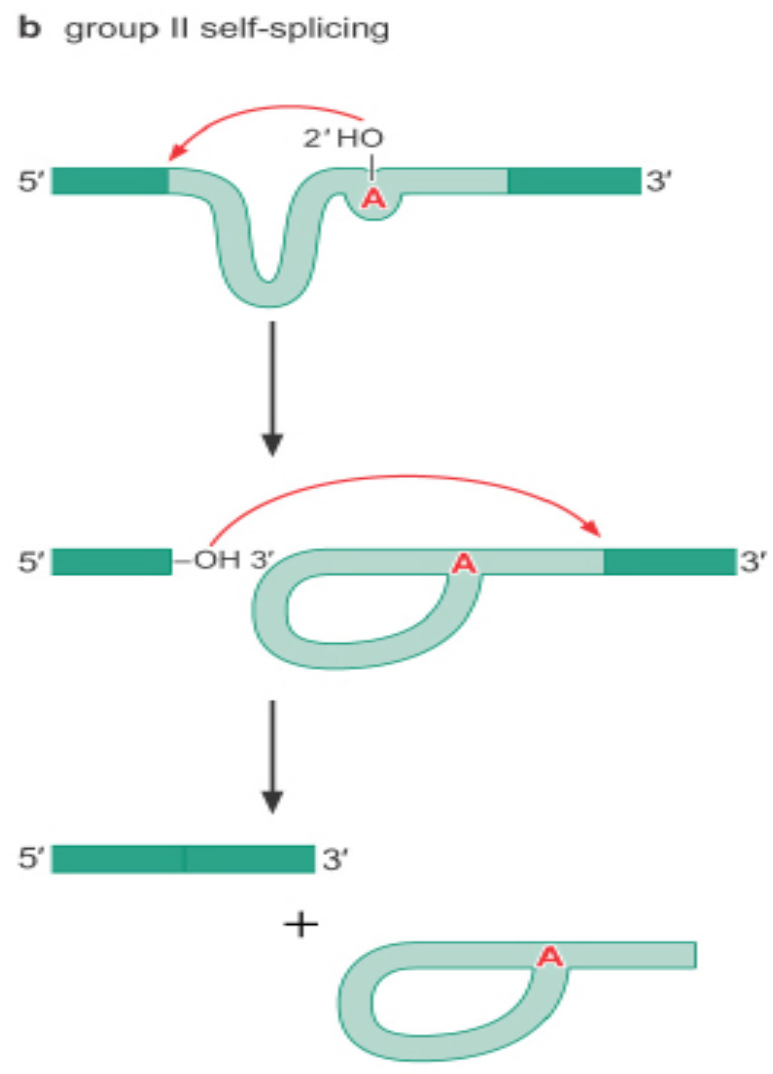


Splicing of Group II introns

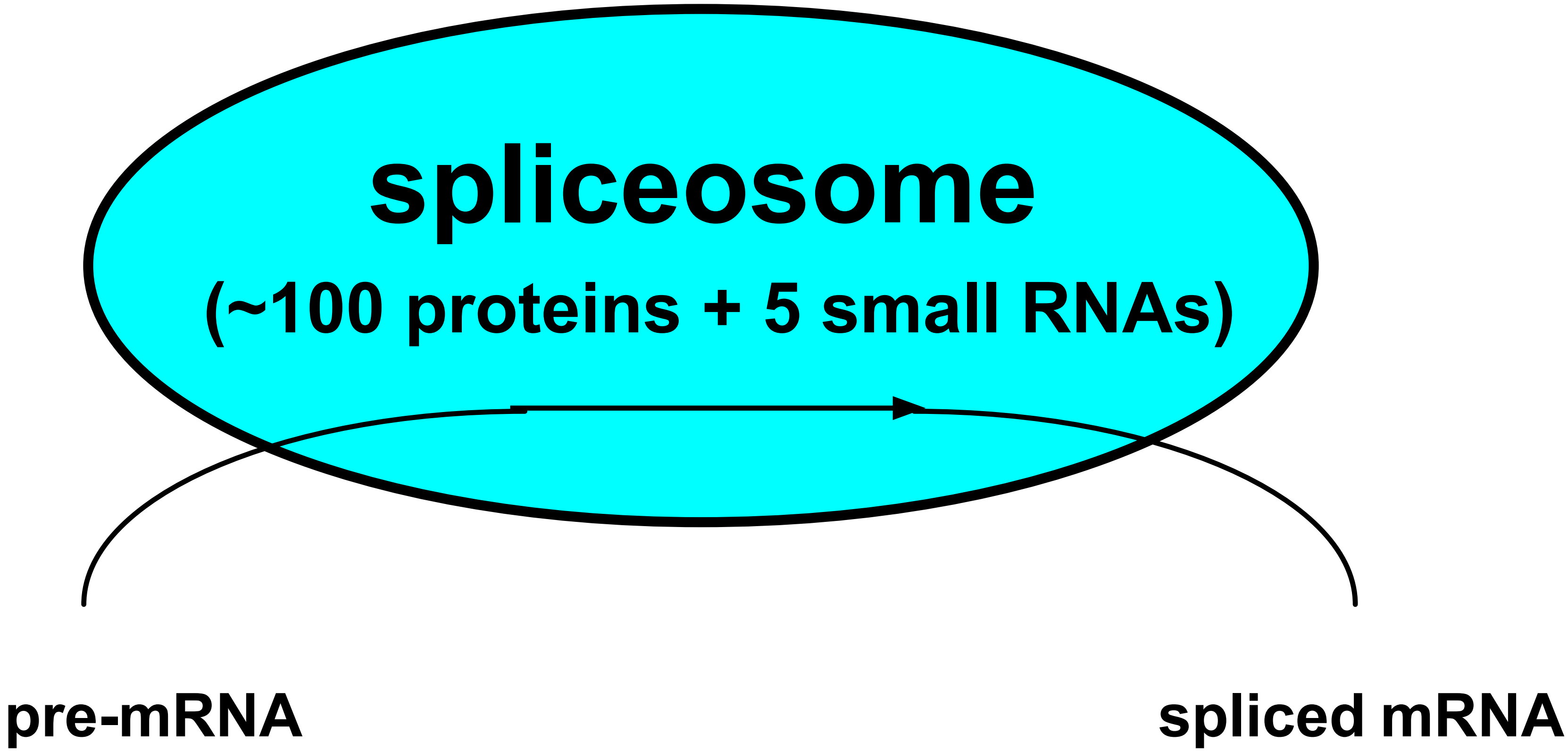
Group II

– self-splicing

- Initiate splicing with an **internal A**
- Uses a phosphoester transfer mechanism
- Does **not** require ATP hydrolysis



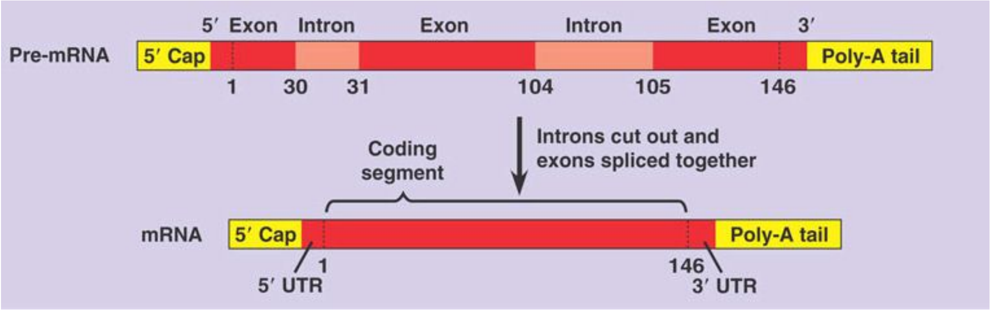
Splicing of pre-mRNA occurs in a "spliceosome" an RNA-protein complex

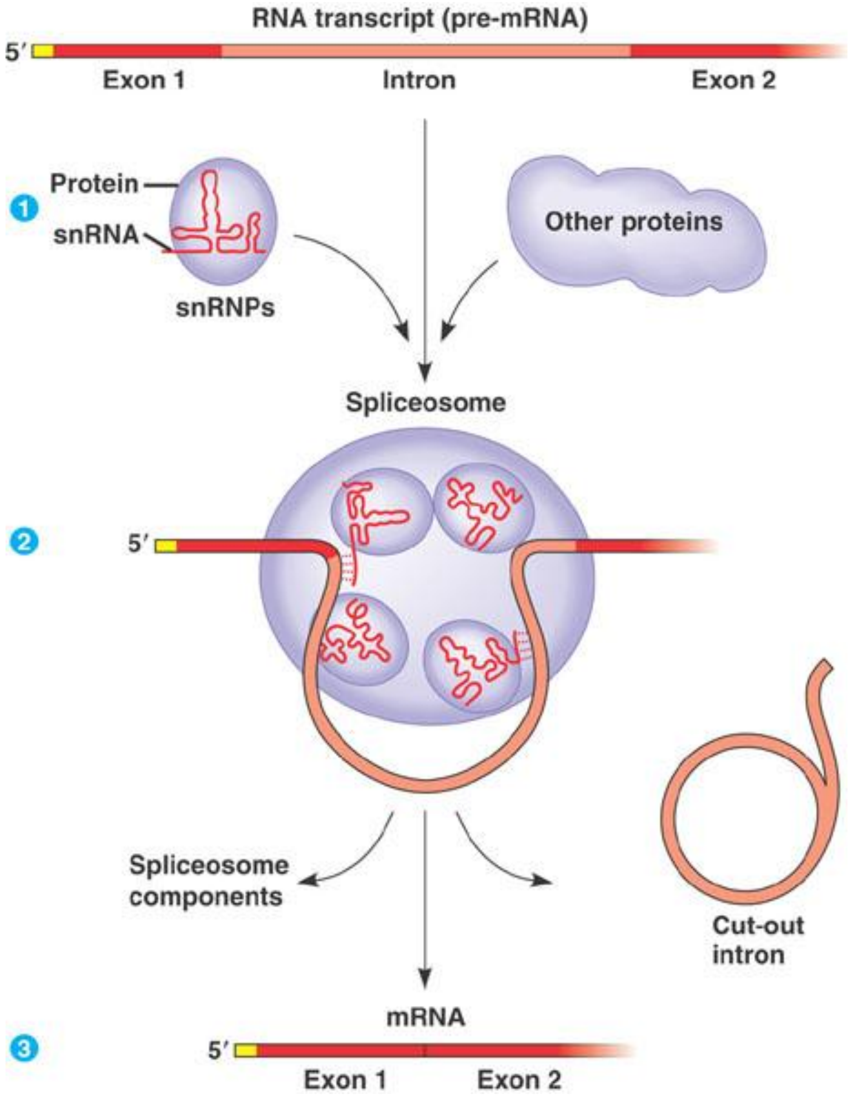


The spliceosome is a large protein-RNA complex in which splicing of pre-mRNAs occurs.

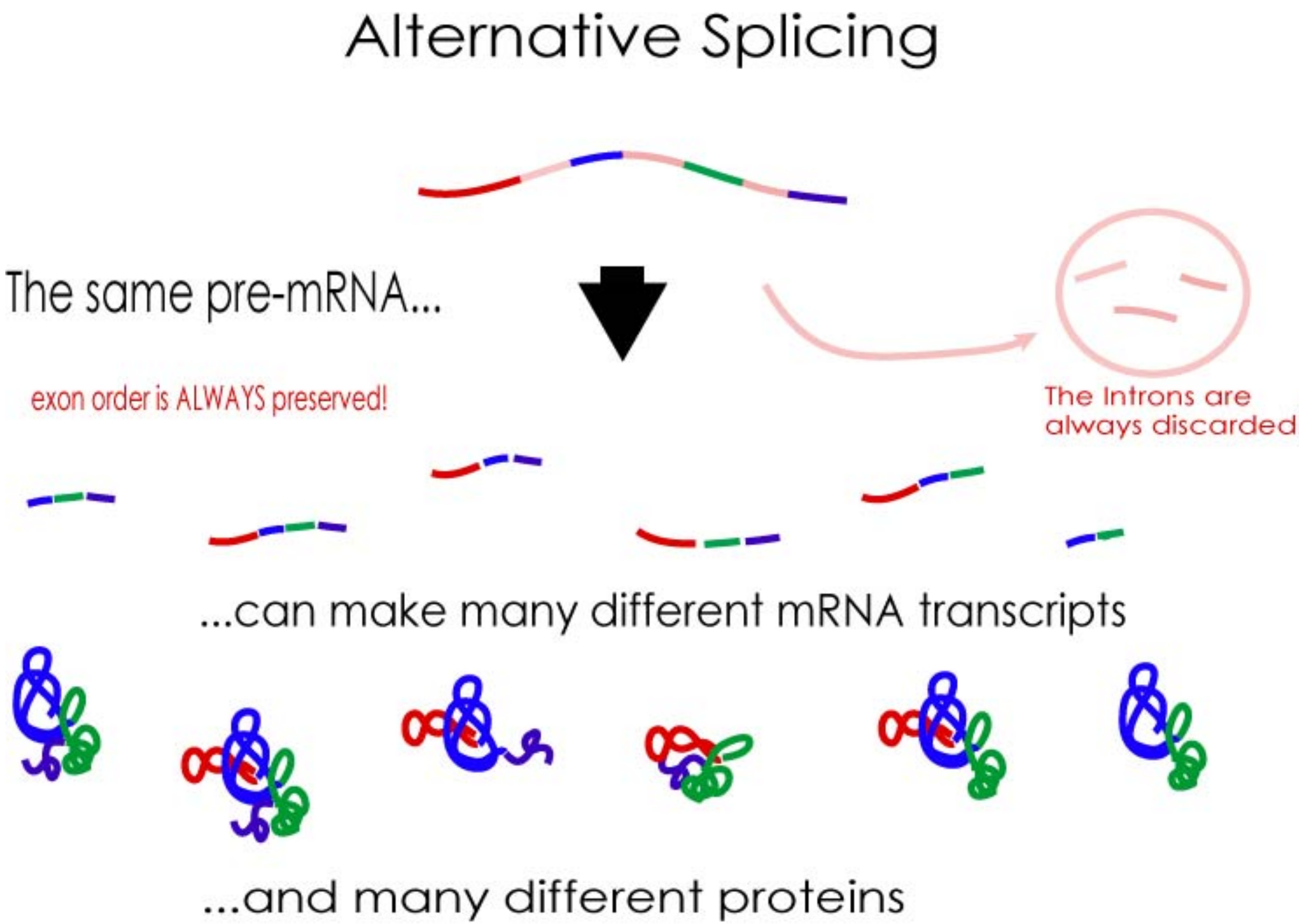
Splicing in Eukaryotes

- Spliceosome of snRNP removes introns, leaves exons





RNA Processing in Eukaryotes



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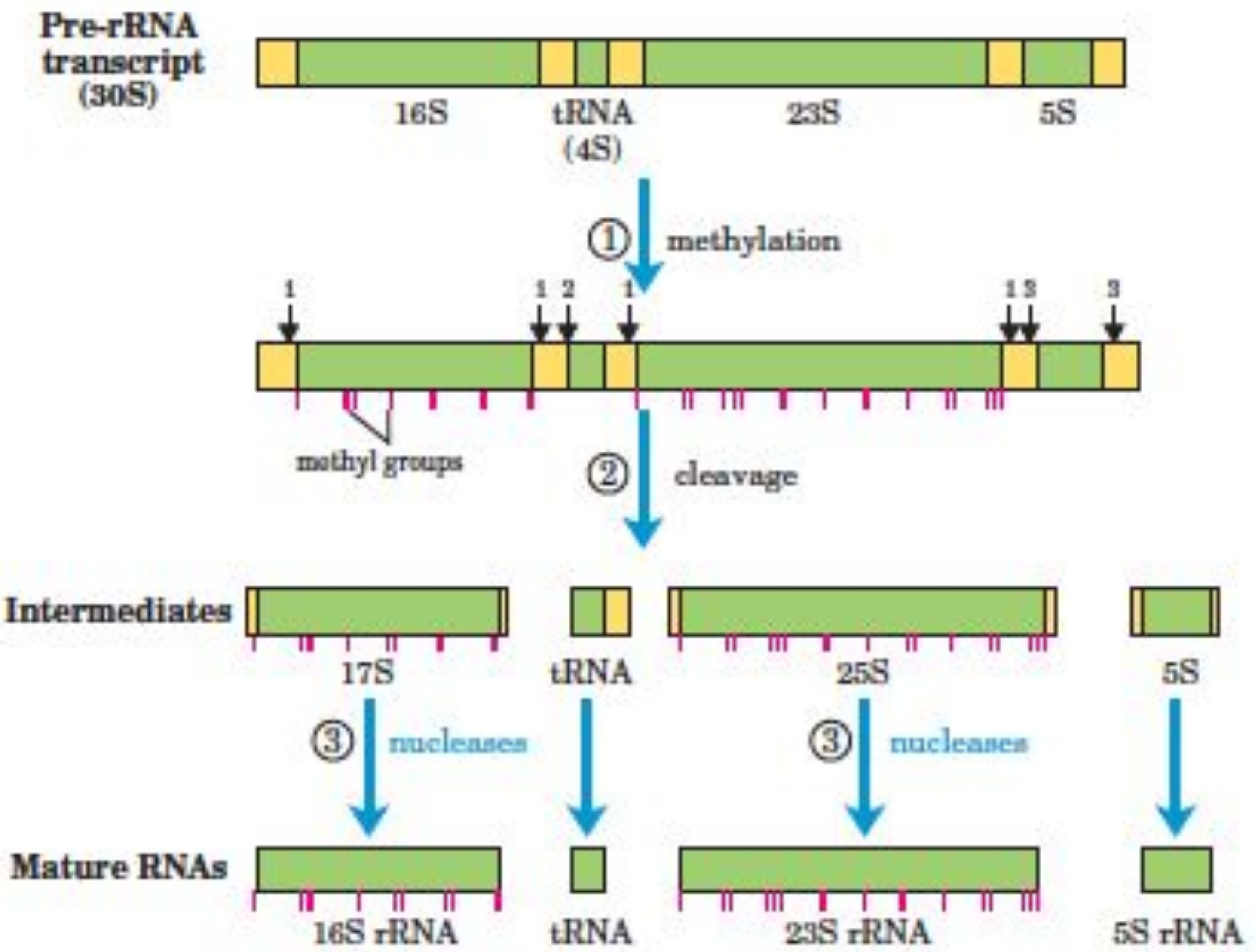
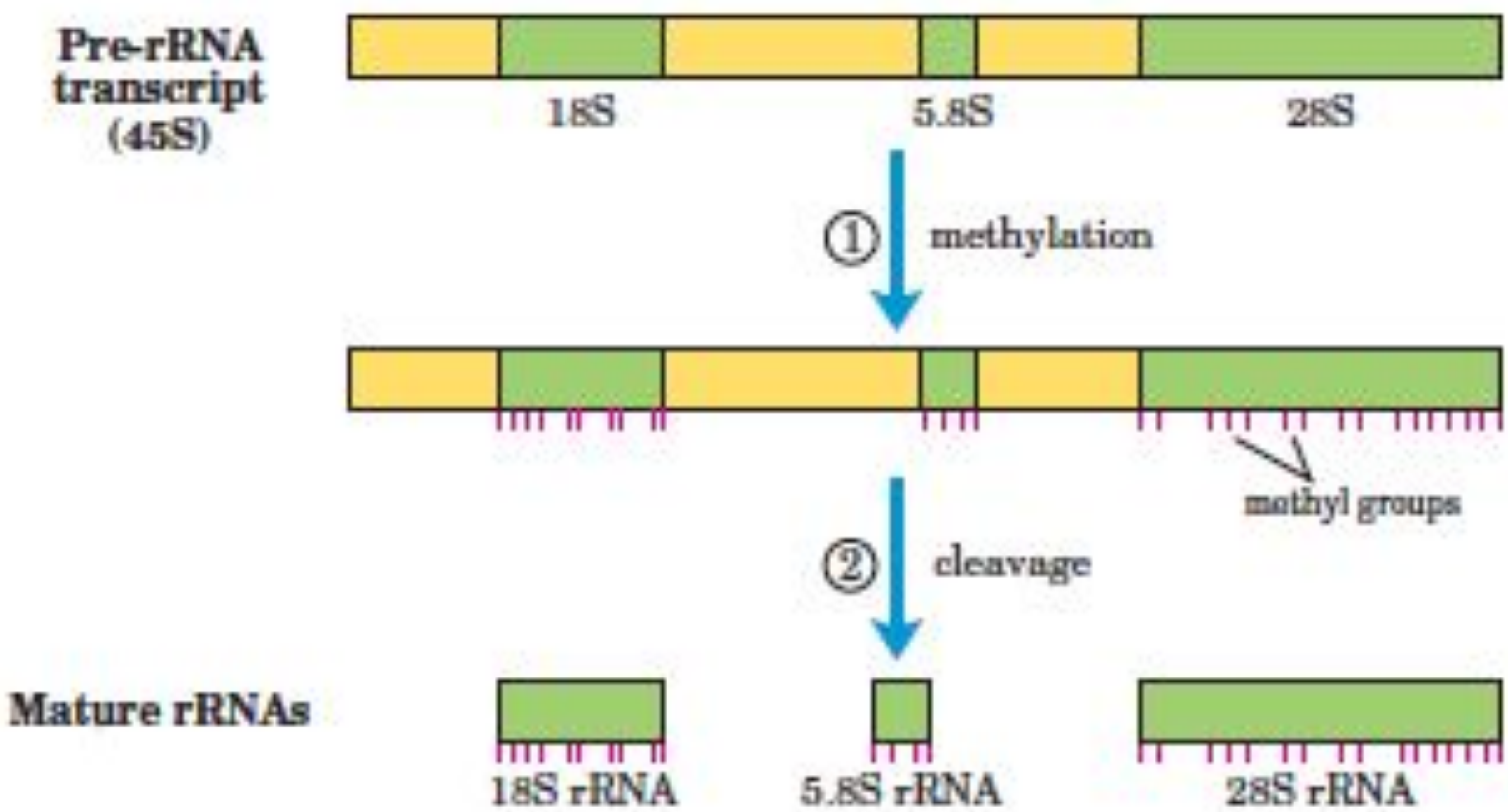
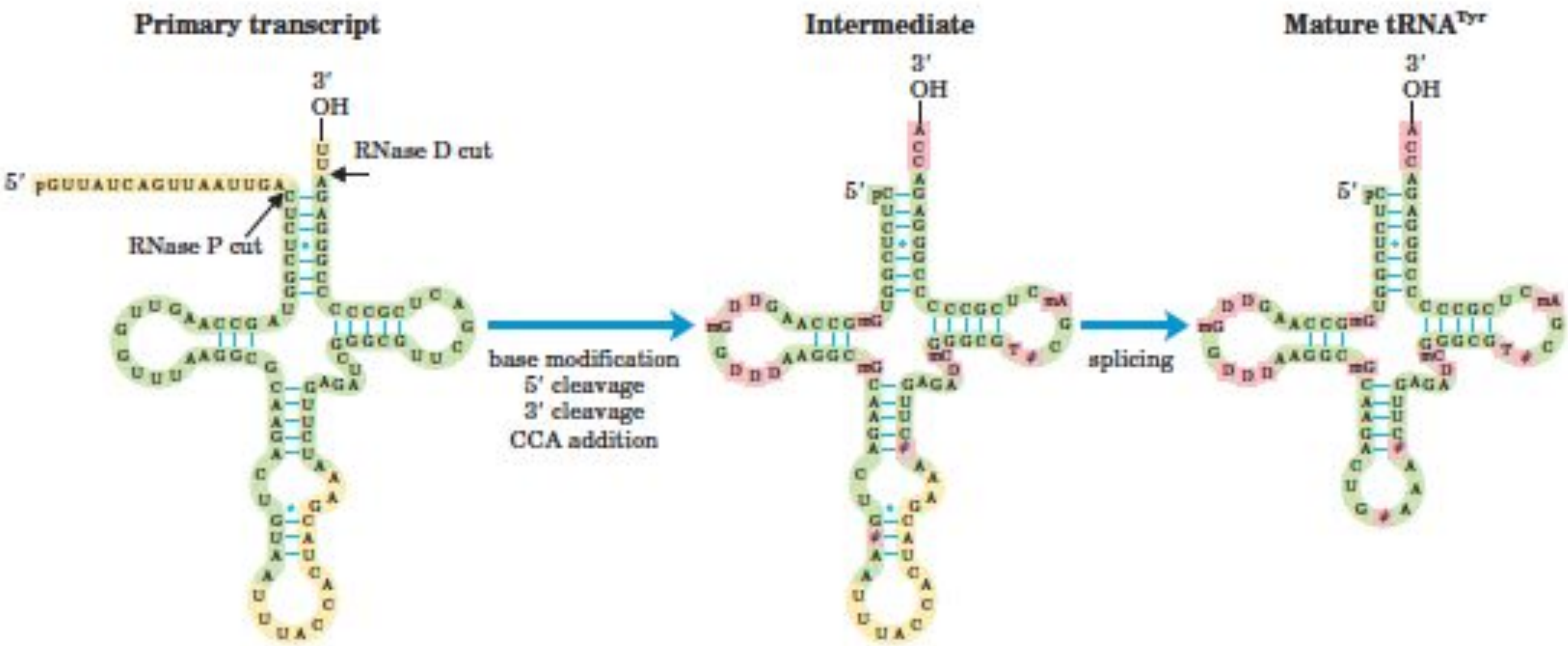
Processing of pre-rRNA transcripts in bacteria



Processing of pre-rRNA transcripts in vertebrates



Processing of tRNAs in bacteria and eukaryotes



Differences between prokaryotic and Eukaryotic Transcription

Sl. No.	Prokaryotic Transcription	Eukaryotic Transcription
1	Transcription and translation are continuous process and occurs simultaneously in the cytoplasm	They are two separate processes, transcription occurs in the nucleus whereas translation occurs in the cytoplasm
2	Transcription initiation machinery is simple since DNA is not associated with any histone proteins	Transcription initiation machinery is very complex since the genetic material is associated with proteins
3	Only one type of RNA polymerase enzyme, which synthesize all types of RNA in the cell (mRNA, rRNA and tRNA)	Three types of RNA polymerase in the cell. RNA Polymerase I for rRNA synthesis RNA Polymerase II for mRNA synthesis. RNA polymerase III for tRNA and 5S rRNA synthesis
4	RNA polymerase with 5 subunits, Two α subunits, One β subunit, One β' subunit, One ω subunit. Functional RNA polymerase is $2\alpha 1\beta 1\beta' \omega$	RNA polymerase I with 14 subunits, RNA polymerase II with 10 -12 subunits, RNA polymerase II with 12 subunits

Differences between prokaryotic and Eukaryotic Transcription

5	σ factor present, which is essential for transcription initiation	σ absent and it is not required for transcription initiation. Initiation of transcription is facilitated by initiation factors
6	RNA polymerase can recognize and bind to the promoter region with the help of σ factor	RNA polymerase cannot recognize the promoter region directly unless the promoter is pre-occupied by transcription initiation factors.
7	Promoter region always located upstream to the start site	Promoter region usually located upstream to the start site, but rarely as in the case of RNA polymerase III, promoter is located downstream to start site
8	Promoter region contain pribnow box at -10 positions. TATA box and CAT box are absent in the promoter region of prokaryotes	Promoter region contains; TATA box located 35 to 25 upstream; CAT box located ~ 70 nucleotide upstream; GC box located ~ 110 nucleotide upstream. Pribnow box absent in eukaryotes

Differences between prokaryotic and Eukaryotic Transcription

9	Termination of transcription is done either by rho dependent mechanisms or rho independent mechanisms	A termination mechanism of transcription is not completely known. It may be direct by the poly A signal or by the presence of termination sequence in the DNA
10	Usually there is no post transcriptional modification of the primary transcript	Primary transcript undergo post transcriptional modifications (RNA editing)
11	RNA capping absent, mRNA is devoid of 5' guanosine cap	RNA capping present, capping occurs at the 5' position of mRNA
12	Poly A tailing of mRNA is absent	Mature mRNA with a poly A tail at the 3' position. Poly A tail is added enzymatically without the complementary strand

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Differences between prokaryotic and Eukaryotic Transcription

13	Introns absent in the mRNA	Introns present in the primary transcript
14	Splicing of mRNA absent since introns are absent	Splicing present, introns in the primary transcript are removed and exons are rejoined by a variety of splicing mechanisms
15	Genes usually polycistronic and hence single transcript may contain sequence for many polypeptides	Genes are monocistronic thus single transcript code for only one polypeptide
16	SD sequence (Shine-Dalgarno sequence) present about 8 nucleotide upstream of start codon in the mRNA, SD sequence act as the ribosome binding site	SD sequence is absent in mRNA of Eukaryotes, but Kozak sequences present in vertebrates

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Inhibitors of DNA-Dependent RNA Polymerase

- Actinomycin D:
  - Inhibit RNA polymerase at elongation by bind with initiation complex in both bacteria and eukaryotes,
  - Act as antibiotic.
- Acridine:
  - Inhibits RNA synthesis in a similar fashion of Actinomycin D.
- Rifampicin:
  - Inhibits bacterial RNA synthesis by binding to the β subunit of bacterial RNA polymerases, preventing the promoter clearance step of transcription,
  - Act as antibiotic.
- α-amanitin:
  - Disrupts mRNA formation in animal cells by blocking Pol II and, at higher concentrations, Pol III.

Reverse Transcription

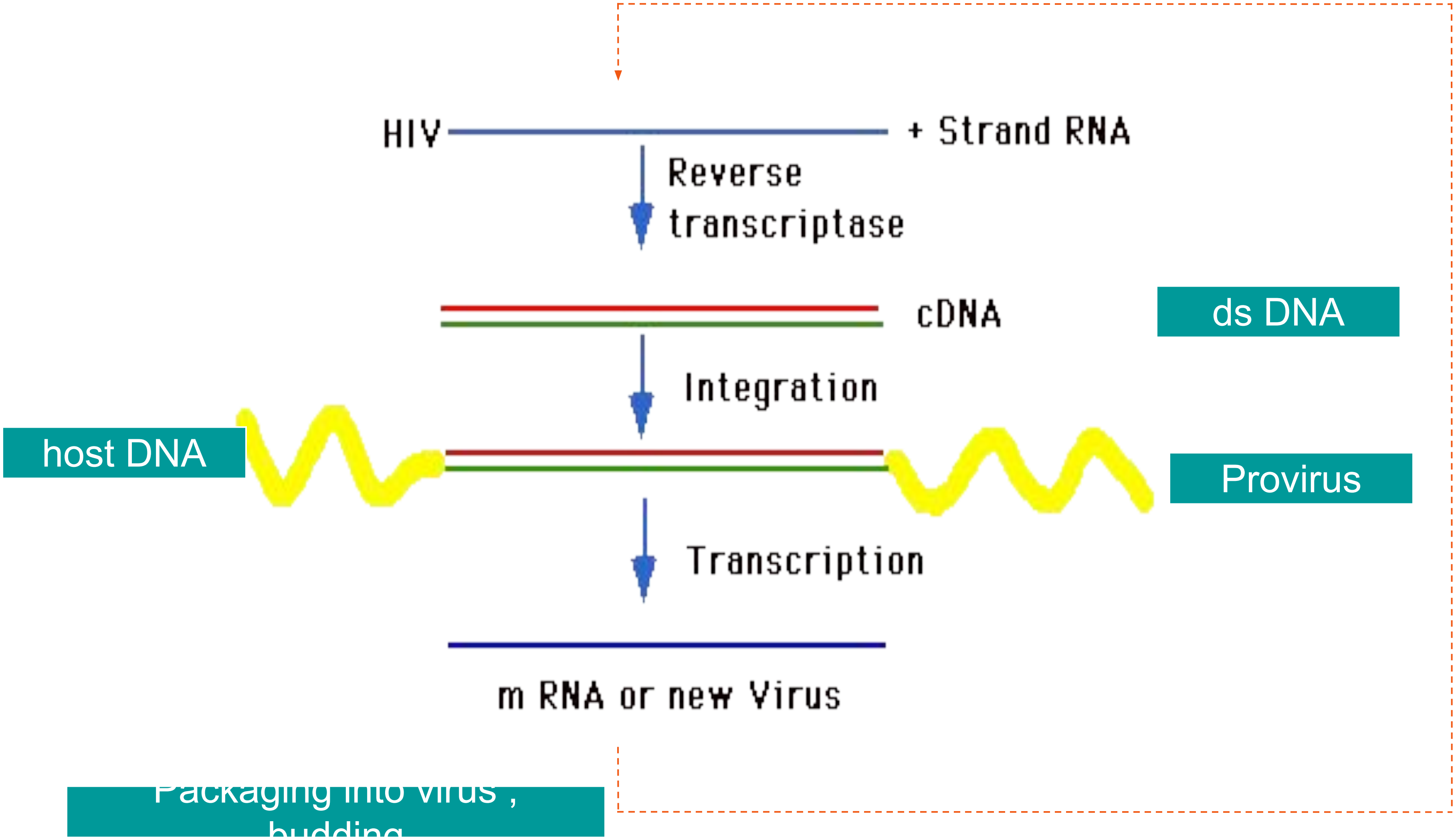
- Normal transcription involves synthesis of RNA from DNA.
- Reverse transcription is the transcription of single stranded RNA into double stranded DNA
- With the help of the enzyme Reverse Transcriptase.
- Reverse Transcriptase also known as
  - RNA directed DNA Polymerase
  - DNA Nucleotidyl transferase (RNA directed)
  - Revertase
- Reverse Transcriptase was discovered by Howard Temin and Baltimore in 1970 independently
  - Shared Nobel Prize in Physiology or Medicine in 1975 for their discovery.

Reverse Transcription

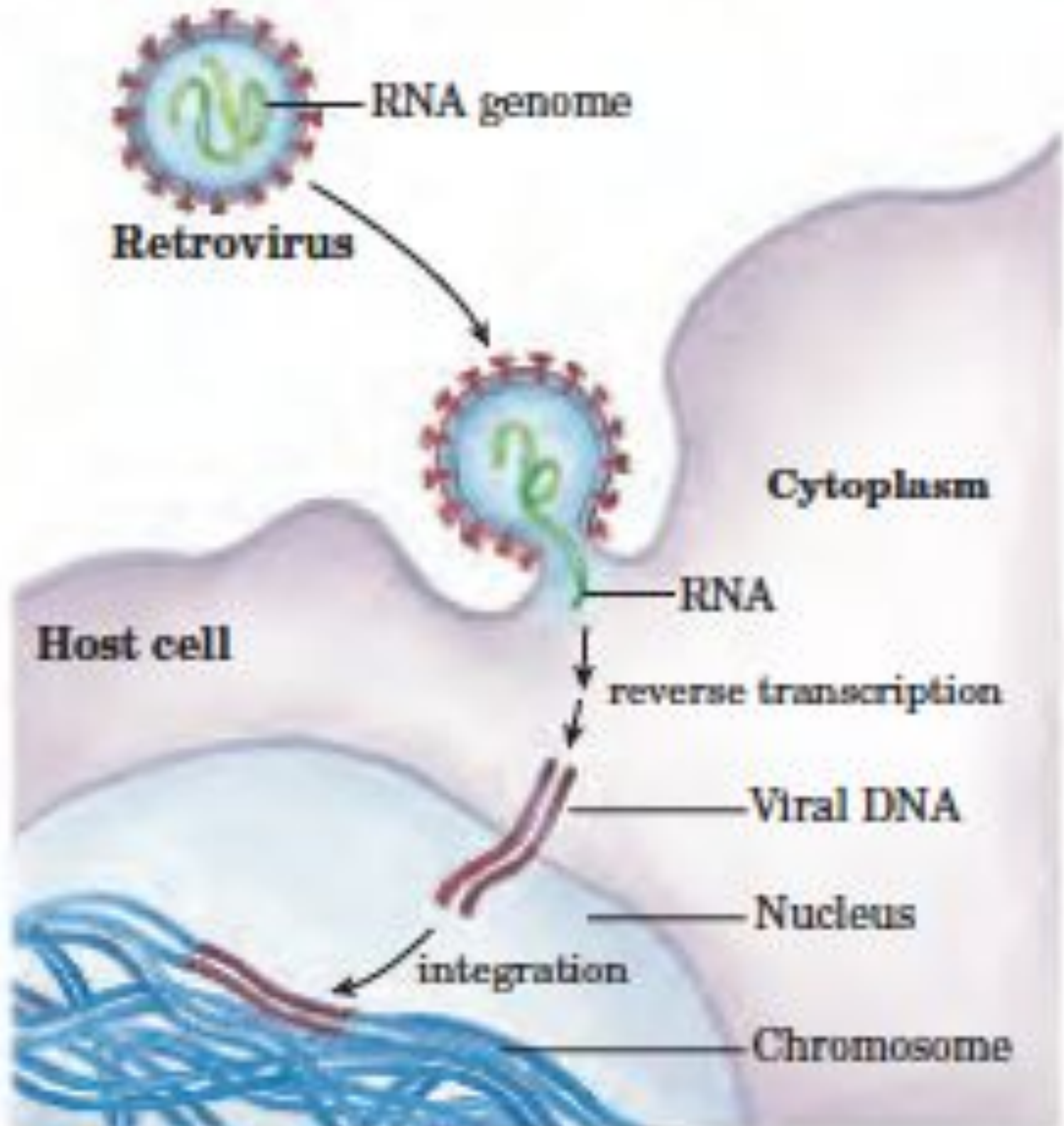
- Reverse transcriptase common in Retrovirus.
  - HIV
  - M-MLV (Moloney Murine Leukemia Virus)
  - AMV (Avian Myeloblastosis Virus)

- Reverse Transcriptase enzyme includes two activity:
  - DNA polymerase
  - RNase H (to hydrolyze RNA strand in an RNA/DNA hybrid)

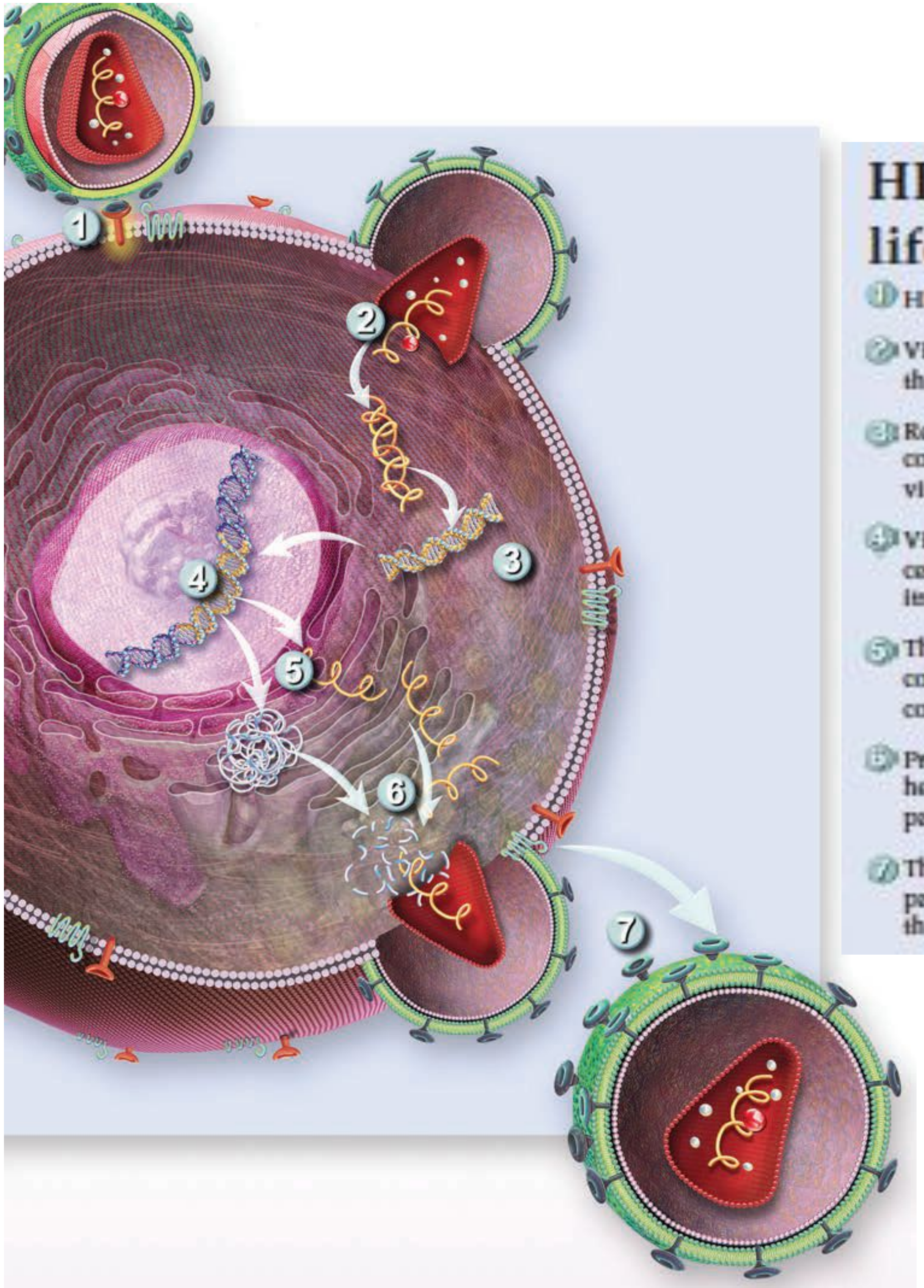
Retrovirus Replication Cycle



Retrovirus Replication Cycle



HIV life Cycle



HIV life cycle

1. HIV binds to the T-cell.
2. Viral RNA is released into the host cell.
3. Reverse transcriptase converts viral RNA into viral DNA.
4. Viral DNA enters the T-cell's nucleus and inserts itself into the T-cell's DNA.
5. The T-cell begins to make copies of the HIV components.
6. Protease (an enzyme) helps create new virus particles.
7. The new HIV virion (virus particle) is released from the T-cell.

Read more….

THANK YOU…